

## Parameter Sensitivity Analysis of Cylindrical $\text{LiFePO}_4$ Battery Performance Using Multi-Physics Modeling

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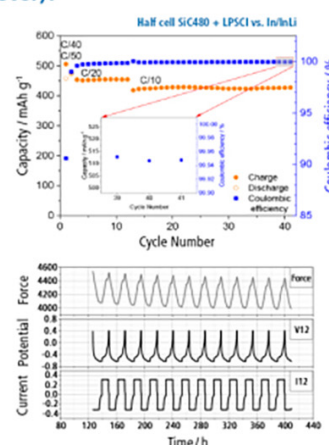
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## Parameter Sensitivity Analysis of Cylindrical LiFePO<sub>4</sub> Battery Performance Using Multi-Physics Modeling

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A multi-physics model for a cylindrical Li-ion battery has been developed by coupling the thermal distribution in the radial direction to an electrochemical P2D model. The model can predict both terminal voltage and surface temperature, which has the advantage that it can be readily validated by measurement. A sensitivity analysis of up to 30 parameters was carried out using model simulation. A parameter sensitivity matrix was established to describe the parameter sensitivity under different operating conditions and the parameters were grouped according to their average sensitivity. The parameters were clustered based on their sensitivity matrix with Fuzzy C-Means (FCM) method. The cluster centers are special operating conditions on which the parameters in the same cluster have high identifiability. Finally, a stepwise experiment is designed based on the analysis results of parameter sensitivity, and the rationality and effectiveness are also validated. It was shown that the stepwise approach to parameter identification results in significantly higher accuracy.

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In recent years, Li-ion batteries have been widely used in electric vehicles due to their relatively high energy and power density. In such demanding applications, Prognostics and Health Management (PHM) of Li-ion batteries is a key issue and has attracted increasing interest from researchers. The aim of our research is to build a PHM database for Li-ion batteries by determining which parameters will change and how they change under different aging patterns. This database could be used for characterization of State-of-Health (SOH) and estimation of remnant lifetime. The core work is to develop a numerical model that can predict battery behavior and identify the critical model parameters accurately and nondestructively using simple and rational experiments.

The Newman P2D model is a type of first principle model commonly used in the analysis of Li-ion battery performance.<sup>1-3</sup> It addresses the complex interaction of physicochemical processes such as diffusion, ion migration, ohmic phenomena and electrochemical reaction using a group of partial differential and algebraic equations. The parameters of the P2D model have corresponding physical meaning and can be treated as an indicator of battery SOH, e.g. Zhang et al. used the stoichiometric number of the electrode material to indicate the stages of capacity fading.<sup>4,5</sup> Schmidt et al. investigated the relationship between the parameters  $\varepsilon_s$ ,  $\kappa_e$  and the cycle number during aging and used them to evaluate the SOH of batteries.<sup>6</sup> Ramadesigan et al. found that the parameters  $k_{s,a}$  and  $D_{s,a}$  degraded with cycle number according to a power-law.<sup>7</sup> These papers have established relationships between a limited number of parameters and battery SOH. The aim of this work was to investigate a much larger set of model parameters under different stages of aging. Therefore, a rapid, accurate and non-invasive identification method is necessary.

However, the large number of parameters in the P2D model makes parameter identification computationally challenging. For example, it takes about three weeks to identify 88 parameters on a cluster of five quad-core PCs using a dynamic test.<sup>8</sup> The identifiability of parameters and identification accuracy are also important issues. Forman et al. used the Fisher information matrix to study parameter identifiability.<sup>9</sup> They concluded that some of the parameters in the P2D model are unidentifiable from experimental data because they do not affect the model output in a unique and significant way. They also summarized the important factors that determine parameter identifiability. Firstly, the identifiability is a function of which parameters are being identified and which are assumed to be already known. Secondly, the identifiability also depends on the values of the parameters after they

have been fitted to the data. Finally, the experimental procedure can greatly influence parameter identifiability.

Identifiability can be indicated by parameter sensitivity, i.e. the influence on battery model output when a parameter changes value. When the value of a highly sensitive parameter changes a little, the model output, i.e. the terminal voltage and surface temperature, will change noticeably. Thus it is easier to identify a highly sensitive parameter from experiments. On the other hand, low sensitivity parameters have very little effect on the model output and are therefore hard to identify correctly from experiments. Insensitive parameters cannot be identified and should be excluded from identification.

Schmidt et al.<sup>10</sup> also used the Fisher information matrix to analyze parameter sensitivity and excluded some unidentifiable parameters to increase the identification quality of the remaining parameters. They also proposed an identification scheme with five different experiments, in which the parameters were grouped and identified stepwise. Using this method the variance and confidence interval can be obtained, but the identifiability matrix needs to be calculated, which is very complex with a multi-physics model. Min et al.<sup>11</sup> and Wang et al.<sup>12</sup> both used model simulation to examine parameter sensitivity for a PEM fuel cell model. They roughly classified the parameters into several groups without a quantitative method. Srinivasulu et al. provided a simple method based on simulated polarization curves to quantify the parameter sensitivity for a PEM fuel cell model.<sup>13</sup> A similar method has not yet been applied to Li-ion batteries. Moreover, there are few established standards for identification experiments; constant current tests<sup>7,14</sup> and dynamic tests<sup>8,10</sup> have typically been used by researchers. These all tend to employ the terminal voltage as the primary model output, while the surface temperature of the battery has been seldom used.

In this article, we analyze the parameter sensitivity of our multi-physics model and propose a feasible method to design experiments for identifying the parameters more accurately using terminal voltage  $U_{app}$  and surface temperature  $T_{sh}$  as two identification objects. The remainder of the paper is organized as follows: Section 2 extends the P2D model to a multi-physics model by adding the thermal behavior and temperature distribution in the radial direction of a cylindrical Li-ion battery. In this way the surface temperature, which is a readily measurable quantity, can be simulated more accurately. Section 3 analyzes the parameter sensitivity and proposes a parameter sensitivity matrix to describe the sensitivity of model output when parameters take different values on different operating conditions. In Section 4, a stepwise experiment is designed based on the parameter sensitivity analysis to facilitate more accurate parameter identification. Validation studies of the rationality and effectiveness of the stepwise experiment are also presented. Section 5 summarizes the main conclusions of the paper.

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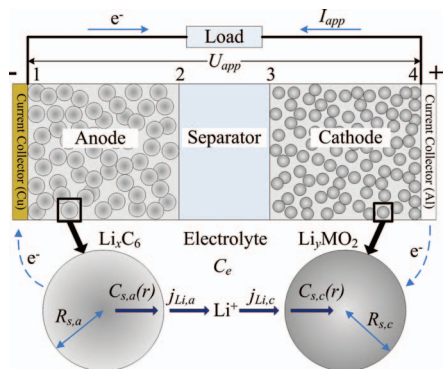
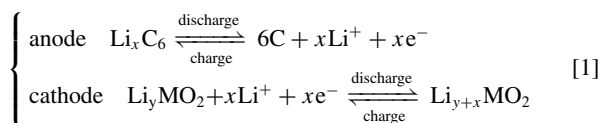


Figure 1. Schematic of P2D model (discharge process).

### Multi-physics Model

The multi-physics model developed in this work consists of the P2D electrochemical model coupled to a thermal model of heat flow and temperature distribution in a cylindrical Li-ion battery as described below.

**Electrochemical model.**— The schematic of the P2D model for a Li-ion battery is shown in Fig. 1. It consists of two current collectors, a negative electrode (anode), a separator and a positive electrode (cathode). The electrodes and separator have a porous structure. Two inner boundaries (anode/separator interface 2 and separator/cathode interface 3) and two external boundaries (Cu/anode interface 1 and cathode/Al interface 4) are also shown. The electrochemical reactions occurring in the electrodes during charge and discharge processes can be expressed as follows:



where  $\text{MO}_2$  stands for metallic oxide, such as  $\text{CoO}_2$  or  $\text{FePO}_4$ , and  $x$  and  $y$  are the stoichiometric numbers of anode and cathode respectively, i.e. the ratio of  $\text{Li}^+$  concentration  $C_s$  and maximum  $\text{Li}^+$  concentration  $C_{s,\text{MAX}}$ .

During the discharge process, active lithium diffuses from within the bulk of the anode material to the surface, where lithium ions ( $\text{Li}^+$ ) are generated and released into the electrolyte by the electrochemical reaction which occurs on the solid/electrolyte interface, releasing an equal number of electrons ( $e^-$ ). Lithium ions move through the electrolyte toward the cathode through the separator, then insert into the cathode active material via another electrochemical reaction. The electrons cannot pass through the separator, so they migrate to the current collectors and the external circuit, thereby providing the electrical work. These physical and chemical processes are described by several partial differential equations and algebraic equations summarized in Appendix A.

On the other hand, thermal phenomena such as energy conservation, heat generation and conduction also occur during the charge/discharge process and can be taken into account in the P2D model. In addition, some model parameters (e.g. diffusion coefficient, conductivity and reaction rate) depend on the battery temperature (see Appendix B for details). A coupled electrochemical-thermal model is presented by feeding back the relationship between temperature-dependent parameters and thermal effects to the P2D model.

**Thermal distribution of cylindrical Li-ion battery.**— A typical cylindrical Li-ion battery consists of a spiral cell as shown in Fig. 2. Thin layers of anode, cathode, separator and current collector are stacked as shown in Section B-B, rolled up on a central mandrel (Section A-A) and then inserted into a can. The gaps are filled with liquid electrolyte.

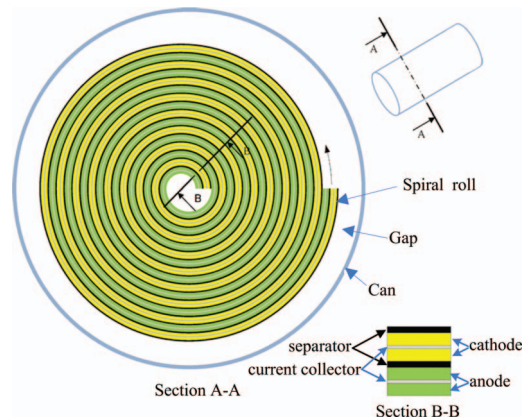


Figure 2. Cross-section of a cylindrical Li-ion battery.<sup>15</sup>

Therefore, thermal distribution occurs in three different directions: (a) the thickness direction of the electrode plate, (b) the radial direction of the cylindrical battery, and (c) the axial direction of the cylindrical battery. The Biot number, which is used to determine whether or not the temperature gradient is negligible, is defined as:

$$Bi = hL/\lambda. \quad [2]$$

where  $h$  is the heat-transfer coefficient,  $L$  is a characteristic length and  $\lambda$  is the effective thermal conductivity. If the Biot number is greater than 0.1, the temperature gradient in that direction cannot be neglected.<sup>16</sup>

Generally, the thickness of the electrode plate in a small battery is several hundred microns, while the heat transfer coefficient and the thermal conductivity of the electrode material are of the same order of magnitude, so the Biot number is usually well below 0.1 and therefore the temperature distribution in the through-thickness direction can be neglected. The temperature distribution in the axial direction can be also be neglected because the axial thermal conductivity  $\lambda_a$  is far higher than the radial thermal conductivity  $\lambda_r$ . The ratio  $\lambda_a/\lambda_r$  is more than 100<sup>17</sup> and the characteristic length is equal to the battery height which is just a few centimeters, yielding a Biot number less than 0.1. However, for the spiral roll configuration, the heat-transfer coefficient  $h$  is typically one or two orders of magnitude greater than the radial thermal conductivity  $\lambda_r$ , which is limited by large contact thermal resistances between the two different layers of battery materials. For example, the radius of a 26650 type Li-ion battery is 0.013 m, the heat-transfer coefficient of the lateral surface is  $\sim 10\text{--}50 \text{ W m}^{-2} \text{ K}^{-1}$  and the radial thermal conductivity is  $0.2 \text{ W m}^{-1} \text{ K}^{-1}$ , so the Biot number is greater than 0.1. In this case, temperature gradients in the radial direction must be taken into account or the simulated temperature will be very different from the experimental value measured at the surface of the battery.

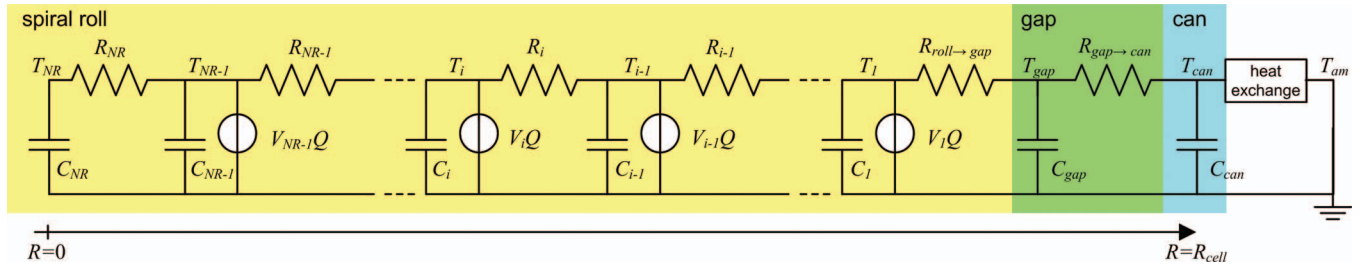
By neglecting the temperature gradient in the thickness direction of the electrode plate and in the axial direction of a cylindrical battery, the heat conduction term in Equation B1 can be reduced to one-dimensional form by Fourier's Law:

$$\nabla(\lambda \nabla T) = \lambda \left( \frac{\partial^2 T}{\partial R^2} + \frac{1}{R} \frac{\partial T}{\partial R} \right) \quad [3]$$

The temperature gradient at the center of the battery is zero and it is equal to the heat exchange rate at the surface:

$$\begin{cases} \frac{\partial T}{\partial R} \Big|_{R=0} = 0 \\ -\lambda \frac{\partial T}{\partial R} \Big|_{R=R_{\text{cell}}} = \dot{q} \end{cases} \quad [4]$$

where  $R_{\text{cell}}$  is the radius of the cylindrical battery and the heat exchange rate  $\dot{q}$  is given by  $\dot{q} = \dot{q}_c + \dot{q}_r$ , where  $\dot{q}_c$  and  $\dot{q}_r$  are the heat exchange rates due to conduction and radiation respectively. The temperature gradient is continuous at the interfaces of spiral roll, gap and can.



**Figure 3.** 1-D thermal impedance model in radial direction of cylindrical battery.

A thermal impedance model based on that presented by Fleckenstein et al.<sup>17</sup> is used to simulate the thermal distribution in the radial direction of a cylindrical Li-ion battery, as shown in Fig. 3.

In Fig. 3, the spiral roll is divided into  $NR$  grids in the radial direction using the finite volume method. Each finite volume element is a cylindrical shell and the innermost one is a cylinder. The volume  $V_i$  and centroid radius  $r_i$  of each element are obtained from the geometric parameters and  $T_i$  is the temperature of the  $i^{th}$  element. The thermal resistance between two adjacent elements is given by:<sup>17</sup>

$$R_i = \frac{1}{\lambda 2\pi H} \ln \frac{r_{i-1}}{r_i} \quad [5]$$

where  $H$  is the effective height of the spiral roll. The thermal capacity is determined by:

$$C_i = V_i \rho C_p \quad [6]$$

Here the density  $\rho$  and heat transfer coefficient  $\lambda$  are volume-averaged values for all materials in the spiral roll.

The thermal impedance model can be described by the state equation in Equation 7. It is noticeable that the innermost element is filled only with separator and electrolyte, so there is no heat source in it, while the gap and the can are treated as independent elements.

Here  $C_{gap} = m_{gap} C_{p,gap}$  is the heat capacity of the gap,  $C_{can} = m_{can} C_{p,can}$  is the heat capacity of the can,  $R_{roll \rightarrow gap}$  and  $R_{gap \rightarrow can}$  represent the thermal resistances between spiral roll, gap and can, calculated in a similar manner to Equation 5,  $A_{sh} \dot{q}$  is the heat exchange between battery shell and the external environment and  $A_{sh}$  is the lateral surface area of the cylindrical battery. The discrete form of the temperature state quantity  $\hat{T}_i$  is  $\hat{T}_i = \frac{T_i^{t+1} - T_i^t}{\Delta t}$ ; when  $t = 0$ ,  $T_i = T_{gap} = T_{can} = T_{am}$ . The temperature distribution  $T_i(t)$  during the charge/discharge process can be calculated by iteration with time step  $\Delta t$ .

The volume averaged value of  $T_i$  to  $T_{NR}$  is used as  $T$  in electrochemical-thermal coupled model. In this work, the temperature differences between the inner/outer grid and the averaged value is not obvious, using the average temperature to update the temperature-related parameters in the electrochemical model may lead to little identification offsets which could be ignored. So, the presented multi-physics model is suitable for small cylindrical battery in our research. When it is used for a large battery, especially in larger current rate profile, the temperature difference may be very obvious. In this case, the whole cylindrical battery should be equivalent to several parallel cells, which have same parameters but different temperature, and the temperature-related parameters should be updated in each equivalent cell, respectively. The simulation of the equivalent cells will be computationally expensive.

$$\begin{bmatrix} \dot{T}_{NR} \\ \dot{T}_{NR-1} \\ \vdots \\ \dot{T}_i \\ \vdots \\ \dot{T}_1 \\ \dot{T}_{gap} \\ \dot{T}_{can} \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_{NR}C_{NR}} & \frac{1}{R_{NR}C_{NR}} & & & & & & \\ \frac{1}{R_{NR}C_{NR-1}} & -\frac{R_{NR}+R_{NR-1}}{C_{NR-1}R_{NR}R_{NR-1}} & \frac{1}{R_{NR-1}C_{NR-1}} & & & & & \\ & & \ddots & & & & & \\ & & & \frac{1}{R_{i+1}C_i} & -\frac{R_{i+1}+R_i}{C_i R_{i+1} R_i} & \frac{1}{R_i C_i} & & \\ & & & & \ddots & & & \\ & & & & & \frac{1}{R_2 C_1} & -\frac{R_2+R_{roll \rightarrow gap}}{C_1 R_2 R_{roll \rightarrow gap}} & \frac{1}{R_{roll \rightarrow gap} C_1} \\ & & & & & \frac{1}{R_{roll \rightarrow gap} C_{gap}} & -\frac{R_{roll \rightarrow gap}+R_{gap \rightarrow can}}{C_{gap} R_{roll \rightarrow gap} R_{gap \rightarrow can}} & \frac{1}{R_{gap \rightarrow can} C_{gap}} \\ & & & & & & \frac{1}{R_{gap \rightarrow can} C_{can}} & -\frac{1}{R_{gap \rightarrow can} C_{can}} \end{bmatrix} \times \begin{bmatrix} T_{NR} \\ T_{NR-1} \\ \vdots \\ T_i \\ \vdots \\ T_1 \\ T_{gap} \\ T_{can} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{V_{NR-1} \dot{Q}}{C_{NR-1}} \\ \vdots \\ \frac{V_i \dot{Q}}{C_i} \\ \vdots \\ \frac{V_1 \dot{Q}}{C_1} \\ 0 \\ -\frac{A_{sh} \dot{q}}{C_{can}} \end{bmatrix} \quad [7]$$



**Table I. Physical and chemical parameters for a 2.3 Ah LiFePO<sub>4</sub> battery.**

| Parameters                                                    | Anode   | Separator | Cathode  | Ref.         |
|---------------------------------------------------------------|---------|-----------|----------|--------------|
| Design specifications                                         |         |           |          |              |
| $A_{cell}$ (m <sup>2</sup> )                                  | 0.1755  | -         | 0.1694   | <sup>b</sup> |
| $L$ (μm)                                                      | 34      | 30        | 70       | <sup>b</sup> |
| $\delta$ (μm)                                                 | 25      | -         | 25       |              |
| $R_s$ (μm)                                                    | 3.5     | -         | 0.0365   | <sup>b</sup> |
| $\epsilon_s$                                                  | 0.55    | -         | 0.43     | <sup>a</sup> |
| $\epsilon_e$                                                  | 0.33    | 0.54      | 0.332    | <sup>a</sup> |
| Li-ion concentrations                                         |         |           |          |              |
| $C_s^{MAX}$ (mol m <sup>-3</sup> )                            | 31370   | -         | 22806    | <sup>b</sup> |
| $C_s^0 / C_s^{MAX}$                                           | 0.8     | -         | 0.03     | <sup>b</sup> |
| Diffusion coefficient and conductivity                        |         |           |          |              |
| $D_s$ (m <sup>2</sup> s <sup>-1</sup> )                       | 3.9e-14 | -         | 1.18e-18 | <sup>a</sup> |
| $\bar{E}D_s$ (KJ mol <sup>-1</sup> )                          | 35      | -         | 35       | <sup>a</sup> |
| $\sigma_s$ (S m <sup>-1</sup> )                               | 100     | -         | 0.5      | <sup>a</sup> |
| $R_{film}$ (ohm m <sup>2</sup> )                              | 0.003   | -         | 0        | <sup>e</sup> |
| Kinetic                                                       |         |           |          |              |
| $k_s$ (m <sup>2.5</sup> mol <sup>-0.5</sup> s <sup>-1</sup> ) | 3e-11   | -         | 1.4e-12  | <sup>a</sup> |
| $\bar{E}k_s$ (KJ mol <sup>-1</sup> )                          | 20      | -         | 30       | <sup>a</sup> |
| $\alpha$                                                      | 0.5     | -         | 0.5      | <sup>b</sup> |
| $\gamma$                                                      | 1.5     | 1.5       | 1.5      | <sup>a</sup> |
| Electrolyte properties                                        |         |           |          |              |
| $C_e$ (mol m <sup>-3</sup> )                                  | 1200    |           |          | <sup>a</sup> |
| $D_e$ (m <sup>2</sup> s <sup>-1</sup> )                       | 2.0e-10 |           |          | <sup>d</sup> |
| $\bar{E}D_e$ (KJ mol <sup>-1</sup> )                          | 11      |           |          | <sup>d</sup> |
| $\kappa_e$ (S m <sup>-1</sup> )                               | 0.2     |           |          | <sup>c</sup> |
| $\bar{E}\kappa_e$ (KJ mol <sup>-1</sup> )                     | 26.6    |           |          | <sup>d</sup> |
| Other parameters and constant                                 |         |           |          |              |
| $t_0^+$                                                       | 0.363   |           |          | <sup>a</sup> |
| $R_{ext}$ (Ω m <sup>2</sup> )                                 | 0.002   |           |          | <sup>e</sup> |
| $R$                                                           | 8.3143  |           |          |              |
| $F$ (C mol <sup>-1</sup> )                                    | 96487   |           |          |              |
| $T_{ref}$ (K)                                                 | 298     |           |          |              |

<sup>a</sup>Ref. 18.<sup>b</sup>Ref. 20.<sup>c</sup>Estimate value from Ref. 3.<sup>d</sup>Ref. 21.<sup>e</sup>Estimate value.

**Parameter set and simulation.**— A set of physical and chemical parameters for a cylindrical 2.3 Ah LiFePO<sub>4</sub> battery is shown in Table I. In the literature, several expressions have been used to describe the temperature dependence of  $D_e$ ,  $\kappa_e$  and  $D_s$ . Considering that these expressions will probably be no longer valid following those specific properties when electrolyte and electrode degradation, we have opted to use a reference value and the Arrhenius law instead. These parameters are treated as unknowns to be analyzed and identified further.

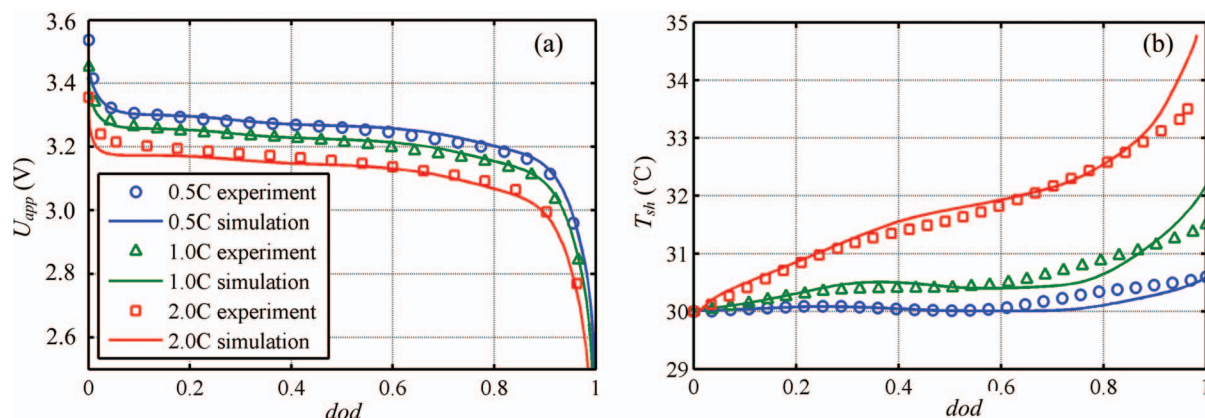
Table II shows the thermal and geometrical parameters of the battery. The thermal properties of the spiral roll are volume-averaged values for all the materials, except the thermal conductivity which was experimentally measured.<sup>18</sup>

We improved DUALFOIL, a freely available FORTRAN code maintained by Newman's research group,<sup>19</sup> by adding subroutines for local heat generation, heat exchange, temperature distribution and parameter updates. This program can simulate both internal behavior (e.g. concentration distribution of electrolyte and reaction current distribution, etc.) and external performance (e.g. terminal voltage and battery temperature) when inputting a set of parameters and an operating condition. It takes about 8 seconds to simulate the 1C discharge process on a PC with INTEL Core i3–530 CPU (2.93 GHz) and 4G RAM. The simulation time is relevant to the operating condition time and will become slower when convergence is an issue, because in this case the step size will become smaller automatically according to the variable step algorithm.

**Table II. Thermal properties and geometrical parameters for a 2.3 Ah LiFePO<sub>4</sub> battery.**

| Thermal properties                                             | $\lambda$ (W m <sup>-1</sup> K <sup>-1</sup> ) | $\rho$ (kg m <sup>-3</sup> ) | $C_p$ (J kg <sup>-1</sup> K <sup>-1</sup> ) |
|----------------------------------------------------------------|------------------------------------------------|------------------------------|---------------------------------------------|
| spiral roll                                                    | 0.2 <sup>a</sup>                               | 2189 <sup>c</sup>            | 1014 <sup>a</sup>                           |
| Electrolyte                                                    | 0.6 <sup>a</sup>                               | 1130 <sup>a</sup>            | 2055 <sup>a</sup>                           |
| can                                                            | 14 <sup>b</sup>                                | 7917 <sup>b</sup>            | 460 <sup>b</sup>                            |
| Cylindrical battery geometry parameters                        |                                                |                              |                                             |
| Diameter (m)                                                   | 26e-3 <sup>b</sup>                             |                              |                                             |
| Height (m)                                                     | 65e-3 <sup>b</sup>                             |                              |                                             |
| Height of roll (m)                                             | 57.2e-3 <sup>b</sup>                           |                              |                                             |
| Diameter of roll (m)                                           | 24.8e-3 <sup>c</sup>                           |                              |                                             |
| mass of roll (kg)                                              | 60e-3 <sup>c</sup>                             |                              |                                             |
| Can thickness (m)                                              | 0.3e-3 <sup>b</sup>                            |                              |                                             |
| mass of can (kg)                                               | 13e-3 <sup>c</sup>                             |                              |                                             |
| heat exchange parameters                                       |                                                |                              |                                             |
| heat transfer coefficient (W m <sup>-2</sup> K <sup>-1</sup> ) | 10 <sup>c</sup>                                |                              |                                             |
| Surface emissivity (W m <sup>-2</sup> K <sup>-4</sup> )        | 0.8 <sup>b</sup>                               |                              |                                             |

<sup>a</sup>Ref. 18.<sup>b</sup>Ref. 15.<sup>c</sup>Estimate value.



**Figure 4.** Comparison of simulated and experimental data at 30°C.

Fig. 4 shows a comparison of simulated discharge curves for 0.5C, 1.0C and 2.0C at 30°C. Simulated curves are represented by solid lines and relevant experimental data by symbols. Reasonable agreement is obtained between the simulated  $U_{app}$  and  $T_{sh}$  curves and the experimental data, with minor differences attributed to the parameters of the experimental battery not being exactly the same as those in Tables I and II. This might be caused by a different production batch or performance degeneration.

Fig. 5 shows the simulated temperature distribution during 2C discharge. The horizontal axis in Fig. 5a indicates the radial direction of a cylindrical Lithium-ion battery, where  $R = 0$  represents the innermost element of the spiral roll and  $R = R_{cell}$  represents the can. There is no heat source in either the gap or the can and the thermal conductivity of the can is very high compared to that of the spiral roll, so a smaller temperature gradient appears there. The shell temperature simulated by our thermal impedance model and the average temperature calculated using an isothermal model, in which the temperatures of all elements in battery are considered the same, are compared with the experimental shell temperature in Fig. 5b. Clearly, the thermal impedance model gives a more accurate prediction of shell temperature, making it possible to identify model parameters by fitting the measured shell temperature to simulation results.

### Parameter Sensitivity Analysis

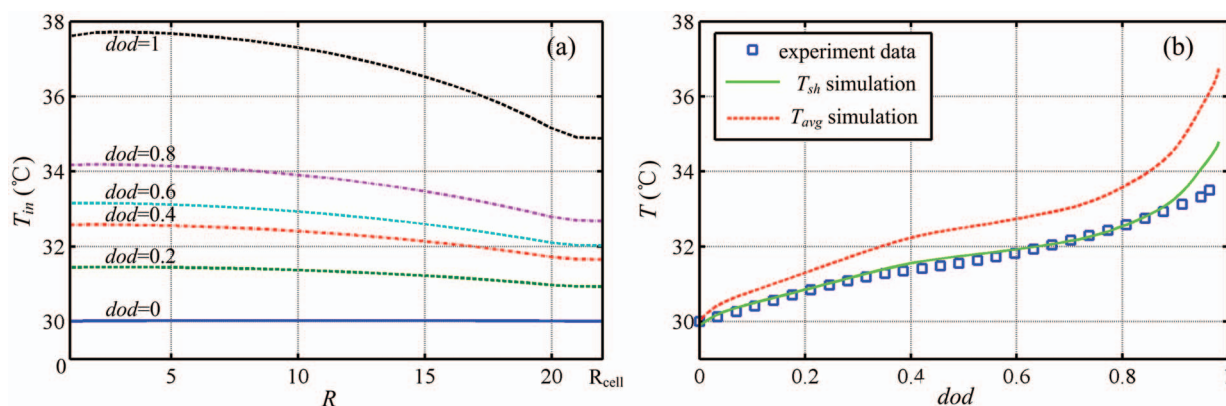
*Parameters to be examined.*— Not all parameters in Table I and II need to be analyzed and identified. Some parameters are unaffected by aging and some can be obtained directly by dismantling the battery, such as the geometrical parameters or material characteristics. A set of

30 parameters, which should be identified by a non-invasive method, is proposed and divided into 2 groups.

Group 1 includes 17 parameters. They are the radii of active particles,  $R_{s,a}$  and  $R_{s,c}$ , the diffusion coefficients in the solid phase,  $D_{s,a}$  and  $D_{s,c}$  and the electronic conductivities in the solid phase,  $\sigma_{s,a}$  and  $\sigma_{s,c}$ . Subscripts a and c refer to anode and cathode respectively. Changes in these parameters indicate degradation of material caused by active particle dissolution and degeneration.<sup>7,22</sup> The initial stoichiometric numbers  $x_0$  and  $y_0$  and the volume fractions of active material  $\varepsilon_{s,a}$  and  $\varepsilon_{s,c}$ , are representative of capacity fade caused by loss of cyclable lithium ions and active material.<sup>4</sup> The film resistances,  $R_{film,a}$  and  $R_{film,c}$ , reflect film formation, which consumes active lithium ions.<sup>5</sup> In addition, the concentration,  $C_e$ , diffusion coefficient,  $D_e$ , ionic conductivity,  $\kappa_e$ , and the density,  $\rho_e$ , of electrolyte will change when reduction reactions and decomposition occur.<sup>23</sup> The external resistance  $R_{ext}$  is also in this group. These parameters characterize the basic properties of the battery, and are most likely to change when the battery degenerates, so we treat them as internal characteristics in the PHM database.

Group 2 includes 13 parameters. They are the electrolyte volume fractions  $\varepsilon_{e,a}$ ,  $\varepsilon_{e,c}$  and  $\varepsilon_{e,s}$ , the reaction rate  $k_{s,a}$  and  $k_{s,c}$ , the activation energy in the Arrhenius law (including  $\bar{E}k_{s,a}$ ,  $\bar{E}k_{s,c}$ ,  $\bar{E}D_{s,a}$ ,  $\bar{E}D_{s,c}$ ,  $\bar{E}D_e$ , and  $\bar{E}\kappa_e$ ), the average thermal conductivity,  $\lambda$ , and the heat transfer coefficient,  $h$ . These parameters may not change as a result of degradation but are difficult to obtain directly.

*Methodology.*— In individual parameter sensitivity analysis, ten equidifferent values for each parameter were chosen from a specific range to analyze their influence on the discharge terminal voltage curve and the shell temperature curve under different conditions with the



**Figure 5.** (a) Radial temperature distribution and (b) comparison of simulated shell temperature, simulated average temperature and experimental shell temperature for cylindrical Li-ion battery during 2C discharge process.

**Table III.** Parameter sensitivity analysis and cluster results.

| Parameters                                                     | Range       | Sensitivity of $U_{app}$ |             |                | Sensitivity of $T_{sh}$ |             |               |            | Clustering results |        |        |
|----------------------------------------------------------------|-------------|--------------------------|-------------|----------------|-------------------------|-------------|---------------|------------|--------------------|--------|--------|
|                                                                |             | $SMU_{avg}$              | $SMU_{max}$ | $Sens.U_{app}$ | $SMT_{avg}$             | $SMT_{max}$ | $Sens.T_{sh}$ | $Sens.tot$ | A                  | B      | C      |
| $R_{s,a}$ ( $\mu\text{m}$ )                                    | 1~9         | 0.0245                   | 0.1401      | ***            | 0.4227                  | 2.1382      | ***           | ***        | 0.1342             | 0.7301 | 0.1356 |
| $R_{s,c}$ ( $\mu\text{m}$ )                                    | 0.02~0.06   | 0.0053                   | 0.0878      | **             | 0.0720                  | 0.7432      | **            | **         | 0.8066             | 0.1094 | 0.0839 |
| $D_{s,a}$ ( $\text{m}^2 \text{s}^{-1}$ )                       | 0.5~10 e-14 | 0.0138                   | 0.1877      | ***            | 0.1064                  | 0.8037      | **            | ***        | 0.2935             | 0.5484 | 0.1579 |
| $D_{s,c}$ ( $\text{m}^2 \text{s}^{-1}$ )                       | 0.5~10 e-18 | 0.0015                   | 0.0471      | *              | 0.0270                  | 0.3648      | *             | *          | 0.7136             | 0.1664 | 0.1199 |
| $\sigma_{s,a}$ ( $\text{S m}^{-1}$ )                           | 10~100      | 1.074e-05                | 3.6285e-05  | /              | 0.0003                  | 0.0016      | /             | /          | /                  | /      | /      |
| $\sigma_{s,c}$ ( $\text{S m}^{-1}$ )                           | 0.1~5       | 0.0052                   | 0.0101      | **             | 0.0712                  | 0.3748      | **            | **         | 0.0803             | 0.8282 | 0.0914 |
| $x_0$ (-)                                                      | 0.75~0.85   | 0.0176                   | 0.0971      | ***            | 0.0610                  | 0.3963      | **            | ***        | 0.1067             | 0.6534 | 0.2397 |
| $y_0$ (-)                                                      | 0.02~0.1    | 0.0019                   | 0.0116      | *              | 0.0422                  | 0.1486      | *             | *          | 0.2885             | 0.4575 | 0.2538 |
| $\varepsilon_{s,a}$ (-)                                        | 0.5~0.6     | 0.0204                   | 0.1317      | ***            | 0.0867                  | 0.6470      | **            | ***        | 0.6011             | 0.2314 | 0.1674 |
| $\varepsilon_{s,c}$ (-)                                        | 0.4~0.5     | 0.0015                   | 0.0096      | *              | 0.0371                  | 0.3103      | *             | *          | 0.1082             | 0.1366 | 0.7551 |
| $R_{film,a}$ ( $\Omega \text{m}^2$ )                           | 0~0.01      | 0.0036                   | 0.0104      | *              | 0.1085                  | 0.5530      | ***           | **         | 0.0600             | 0.0624 | 0.8774 |
| $R_{film,c}$ ( $\Omega \text{m}^2$ )                           | 0~0.01      | 3.026e-05                | 0.0006      | /              | 0.0007                  | 0.0071      | /             | /          | /                  | /      | /      |
| $C_e$ ( $\text{mol m}^{-3}$ )                                  | 800~1200    | 0.0055                   | 0.0071      | **             | 0.0562                  | 0.3118      | **            | **         | 0.5394             | 0.1469 | 0.3135 |
| $D_e$ ( $\text{m}^2 \text{s}^{-1}$ )                           | 0.1~10 e-10 | 0.0062                   | 0.0111      | **             | 0.0578                  | 0.4305      | **            | **         | 0.1002             | 0.1178 | 0.7819 |
| $\kappa_e$ ( $\text{S m}^{-1}$ )                               | 0.4~2       | 0.0150                   | 0.0706      | ***            | 0.1841                  | 2.4130      | **            | ***        | 0.1212             | 0.7164 | 0.1622 |
| $\rho_e$ ( $\text{kg m}^{-3}$ )                                | 1000~1200   | 8.271e-05                | 0.0014      | /              | 0.0155                  | 0.0998      | *             | /          | /                  | /      | /      |
| $R_{ext}$ ( $\Omega \text{m}^2$ )                              | 0~0.002     | 0.0121                   | 0.0351      | ***            | 0.3653                  | 1.8800      | ***           | ***        | 0.8864             | 0.0651 | 0.0484 |
| $\varepsilon_{e,a}$ (-)                                        | 0.3~0.4     | 0.0247                   | 0.1441      | ***            | 0.0811                  | 0.5192      | **            | ***        | 0.1034             | 0.1751 | 0.7214 |
| $\varepsilon_{e,s}$ (-)                                        | 0.5~0.6     | 0.0006                   | 0.0067      | *              | 0.0214                  | 0.1385      | *             | *          | 0.1196             | 0.1688 | 0.7115 |
| $\varepsilon_{e,c}$ (-)                                        | 0.3~0.4     | 0.0013                   | 0.0391      | *              | 0.0191                  | 0.1188      | *             | *          | 0.5396             | 0.2318 | 0.2284 |
| $k_{s,a}$ ( $\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$ ) | 0.1~10 e-11 | 0.0390                   | 0.0579      | ***            | 0.9025                  | 3.0251      | ***           | ***        | 0.5660             | 0.2185 | 0.2154 |
| $k_{s,c}$ ( $\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$ ) | 0.1~10 e-12 | 0.0118                   | 0.0420      | ***            | 0.3348                  | 1.4866      | ***           | ***        | 0.6171             | 0.2701 | 0.1127 |
| $\bar{E}k_{s,a}$ ( $\text{KJ mol}^{-1}$ )                      | 10~50       | 0.0058                   | 0.0202      | **             | 0.1181                  | 0.5533      | **            | **         | 0.8337             | 0.0879 | 0.0782 |
| $\bar{E}k_{s,c}$ ( $\text{KJ mol}^{-1}$ )                      | 10~50       | 0.0013                   | 0.0148      | *              | 0.0384                  | 0.2704      | *             | *          | 0.5068             | 0.1984 | 0.2946 |
| $\bar{E}D_{s,a}$ ( $\text{KJ mol}^{-1}$ )                      | 10~50       | 0.0029                   | 0.0775      | *              | 0.0217                  | 0.4916      | *             | *          | 0.6716             | 0.1555 | 0.1727 |
| $\bar{E}D_{s,c}$ ( $\text{KJ mol}^{-1}$ )                      | 10~50       | 0.0006                   | 0.0091      | *              | 0.0133                  | 0.2449      | *             | *          | 0.5040             | 0.1776 | 0.3183 |
| $\bar{E}D_e$ ( $\text{KJ mol}^{-1}$ )                          | 10~50       | 0.0013                   | 0.0085      | *              | 0.0366                  | 0.3098      | *             | *          | 0.1342             | 0.7301 | 0.1356 |
| $\bar{E}\kappa_e$ ( $\text{KJ mol}^{-1}$ )                     | 10~50       | 0.0112                   | 0.0657      | ***            | 0.2254                  | 1.6048      | ***           | ***        | 0.4335             | 0.2472 | 0.3191 |
| $\lambda$ ( $\text{W m}^{-1} \text{K}^{-1}$ )                  | 0.1~1       | 0.0004                   | 0.0061      | *              | 0.1948                  | 1.6146      | **            | *          | 0.8066             | 0.1094 | 0.0839 |
| $h$ ( $\text{W m}^{-2} \text{K}^{-1}$ )                        | 5~50        | 0.0014                   | 0.0203      | *              | 0.5873                  | 4.4766      | ***           | **         | 0.2935             | 0.5484 | 0.1579 |

\*\*\*Highly sensitive.

\*\*Sensitive.

\*Low sensitive.

/Insensitive.

other parameters equal to benchmark values. The ranges examined for each parameter are shown in Table III, taking into account benchmark values from the literature and the anticipated variation during the aging process. This method has a limitation in that the results are only valid for a given set of benchmark values. If some parameter values are far away from the benchmark value or taken from another battery type or system, the result may be significantly different. In this case, parameter sensitivity should be re-examined in the new range with revised benchmark values.

Simulation of complete discharge of the battery from the fully charged state at five different ambient temperatures and five different discharge rates is shown in Fig. 6. The ambient temperatures ( $T_{am}$ ) are described fuzzily as “Very low”, “Low”, “Room temperature”, “High” and “Very high”, which are  $-5^\circ\text{C}$ ,  $10^\circ\text{C}$ ,  $25^\circ\text{C}$ ,  $40^\circ\text{C}$  and  $55^\circ\text{C}$  according to the batteries’ specification. In a similar way, the discharge current rates ( $C_{rate}$ ) are 0.2C, 0.5C, 1C, 2C and 4C. Thus 25 combined operating conditions are investigated in this paper. Fig. 6 shows the curves for parameters  $R_{s,a}$  and  $\bar{E}k_{s,a}$  as examples.

Different influences of the varied parameters are observed on the simulated  $U_{app}$  and  $T_{sh}$  curves. Both the  $U_{app}$  and  $T_{sh}$  curves show high sensitivity to  $R_{s,a}$  across the entire depth-of-discharge ( $dod$ ) range, as shown in Fig. 6a and 6b, with higher values leading to lower  $U_{app}$ , higher  $T_{sh}$  and a lower End-of-Discharge ( $EOD$ ) voltage. In contrast, varying  $\bar{E}k_{s,a}$  has a more limited influence on the  $U_{app}$  curve in the high  $dod$  region and no influence on  $EOD$ . It is very difficult to characterize parameter sensitivity without a quantitative method, especially with 25 combined operating conditions, and the required method is described below.

**Parameter sensitivity matrix.**— The dispersion of  $U_{app}$  curves indicates the parameter sensitivity of  $U_{app}$ . It was calculated as follows. Firstly, the standard deviation of five  $U_{app}$  values at each  $dod$  point was determined as:

$$SU(dod, j, i) = \text{std}(U_{app}(dod, k, j, i)) \quad [8]$$

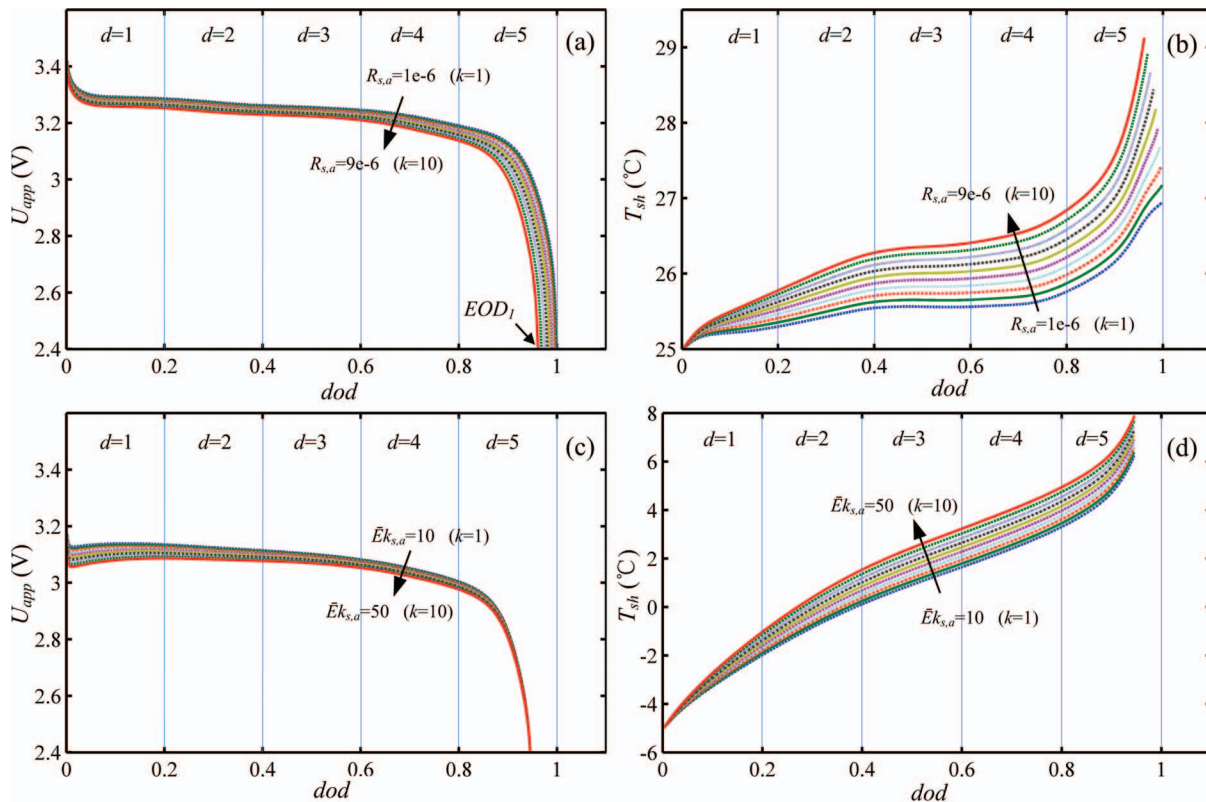
where  $U_{app}(dod, k, j, i)$  represents the terminal voltage at the  $dod^{\text{th}}$  point on the discharge curve for the  $k^{\text{th}}$  value of the parameter, at the  $j^{\text{th}}$  ambient temperature and  $i^{\text{th}}$  current, while  $\text{std}()$  is the standard deviation function.

Secondly, all  $dod$  points were divided into five  $dod$  regions represented by the variable  $d$ , as shown in Fig. 6. The mean values of  $SU(dod, j, i)$  were calculated in each  $dod$  region using Equation 9, where  $EOD_1$  is equal to the  $dod$  at which the curves first reach the  $EOD$  voltage, as indicated in Fig. 6a.

$$\begin{cases} SMU(1, j, i) = \text{mean}_{0 < dod < 0.2} [SU(dod, j, i)] \\ SMU(2, j, i) = \text{mean}_{0.2 < dod < 0.4} [SU(dod, j, i)] \\ SMU(3, j, i) = \text{mean}_{0.4 < dod < 0.6} [SU(dod, j, i)] \\ SMU(4, j, i) = \text{mean}_{0.6 < dod < 0.8} [SU(dod, j, i)] \\ SMU(5, j, i) = \text{mean}_{0.8 < dod < EOD_1} [SU(dod, j, i)] \end{cases} \quad [9]$$

If  $EOD_1$  appears in the region where  $d = 4$ , then:

$$SMU(4, j, i) = SMU(5, j, i) = \text{mean}_{0.6 < dod < EOD_1} [SU(dod, j, i)] \quad [10]$$



**Figure 6.** (a) Terminal voltage curves and (b) Shell temperature curves for various  $R_{s,a}$  at 1C 25°C. (c) Terminal voltage curves and (d) Shell temperature curves for various  $\bar{E}k_{s,a}$  at 2C -5°C.

Finally, a 3-D matrix  $SMU(d,j,i)$  which is termed the “parameter sensitivity matrix” of  $U_{app}$  was obtained using Equations 8 to 10 under 25 permuted conditions, so the dimensionality is (5,5,5). Each element in this matrix represents the parameter sensitivity for a certain  $dod$  region, ambient temperature and current, expressed in units of volts. The  $SMU$  matrices of 30 different parameters were calculated by this method.

The column “ $SMU_{avg}$ ” in Table III shows the mean value of all elements, while the column “ $SMU_{max}$ ” shows the maximum value in the  $SMU$  matrix for each parameter. Parameters whose mean sensitivity was greater than 0.01 were grouped as “Highly Sensitive”, while those with mean sensitivity between 0.005 and 0.01 were grouped as “Sensitive”. “Low Sensitive” parameters were defined as those with mean sensitivity lower than 0.005, but maximum sensitivity greater than 0.005. Parameters with maximum sensitivity lower than 0.005 were grouped as “Insensitive”. This rough grouping is indicated in the “Sens.  $U_{app}$ ” column of Table III.

The  $U_{app}$  parameter sensitivity matrix is shown in Fig. 7 using slice plots, taking  $R_{s,a}$  and  $R_{film,a}$  as examples. It can be seen that the parameter sensitivity is continuously related to operating conditions and  $dod$ .

In the same way, the  $T_{sh}$  parameter sensitivity matrix can be obtained and expressed as  $SMT(d,j,i)$ , with units of degrees Celsius. The mean and maximum sensitivity values of the matrices are shown in the “ $SMT_{avg}$ ” and “ $SMT_{max}$ ” columns in Table III. The parameters were also roughly grouped with threshold values of 0.05 and 0.2 as indicated in the “Sens.  $T_{sh}$ ” column in Table III. The  $SMT$  matrices for parameters  $R_{s,a}$  and  $R_{film,a}$  are shown in Fig. 8.

The  $SMU$  matrix and  $SMT$  matrix for the same parameter show little correlation. Some parameters (e.g. heat transfer coefficient  $h$ ) have a high sensitivity in  $SMT$  but low sensitivity in  $SMU$ . Since temperature tends to increase with depth of discharge,  $T_{sh}$  sensitivity is positively related to  $dod$  under certain operating conditions.  $T_{sh}$  sensitivity is also positively related to current because larger currents

generate more heat and the thermal response is more obvious. The maximum  $SMT$  values of most parameters appear in the high  $dod$  and  $C_{rate}$  regions.

The total sensitivity should take into account both the  $SMU$  and  $SMT$  by summing the number of asterisks in the “Sens. tot” column in Table III. Ten parameters (i.e.  $R_{s,a}$ ,  $D_{s,c}$ ,  $x_0$ ,  $\kappa_e$ ,  $\epsilon_{s,a}$ ,  $R_{ext}$ ,  $\epsilon_{e,a}$ ,  $k_{s,a}$ ,  $k_{s,c}$  and  $\bar{E}k_e$ ) have 5 or 6 asterisks in total, they are regrouped as “Highly Sensitive”. Seven parameters (i.e.  $R_{s,c}$ ,  $\sigma_{s,c}$ ,  $R_{film,a}$ ,  $C_e$ ,  $D_e$ ,  $\bar{E}k_{s,a}$  and  $h$ ) have 4 asterisks in total and they are regrouped as “Sensitive”. Ten parameters ( $D_{s,c}$ ,  $y_0$ ,  $\epsilon_{s,c}$ ,  $\epsilon_{e,s}$ ,  $\epsilon_{e,c}$ ,  $\bar{E}k_{s,c}$ ,  $\bar{E}D_{s,a}$ ,  $\bar{E}D_{s,c}$ ,  $\bar{E}D_e$  and  $\lambda$ ) have 2 or 3 asterisks and they are regrouped as “Low sensitive”. Three parameters ( $\sigma_{s,a}$ ,  $R_{film,c}$  and  $\rho$ ) have less than two asterisks; they are regrouped as “Insensitive” and will be excluded from further analysis, because they have negligible influence on either terminal voltage or surface temperature and the identification results will be not credible and cannot be used for studying the PHM characteristics.

**Clustering analysis.**— Theoretically, a parameter can be identified most easily and accurately under the operating condition that provides the highest sensitivity. This is termed the “Best Conditions for Identification (BCI)” and indicated by the index of the parameter sensitivity matrix. Due to the continuous relationship between parameter sensitivity and operating conditions, conditions similar to BCI could also provide a relatively high sensitivity, while conditions that are opposite to BCI will result in a relatively low sensitivity. For example, the BCI for parameter  $R_{s,a}$  is “low  $T_{am}$ , very high  $C_{rate}$  and very high  $dod$ ”; under these conditions the  $U_{app}$  and  $T_{sh}$  sensitivity are both relatively high. The similar conditions “room temperature, medium  $C_{rate}$ , and medium  $dod$ ” result in a medium sensitivity, while “high  $T_{am}$ , low  $C_{rate}$ , and low  $dod$ ”, which is opposite to BCI provides relatively low sensitivity. This conclusion can be easily observed from inspection of Fig. 7a and Fig. 8a.

The best identification strategy is to identify each parameter under its own BCI or under conditions similar to its BCI, but this is at best



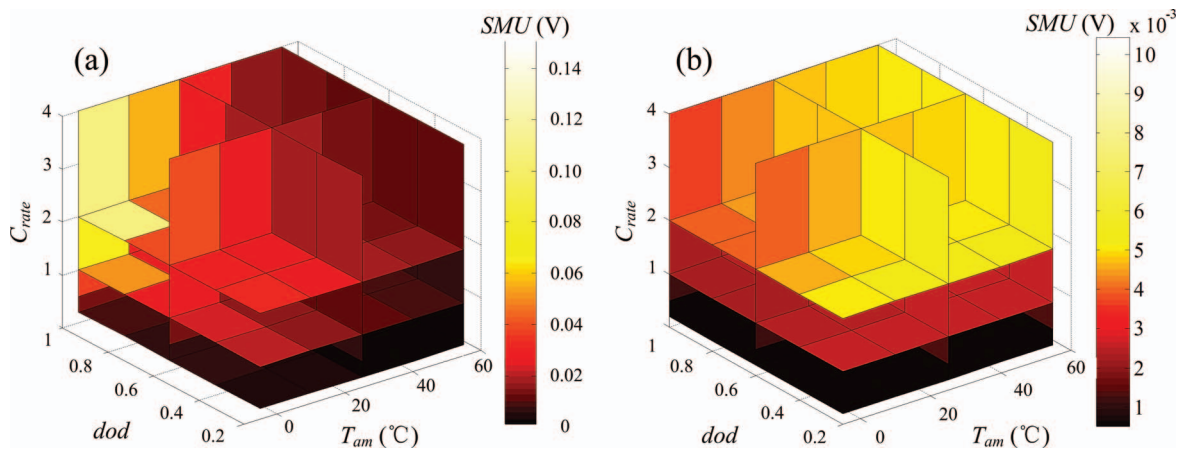


Figure 7.  $U_{app}$  parameter sensitivity matrix for (a)  $R_{s,a}$  and (b)  $R_{film,a}$ .

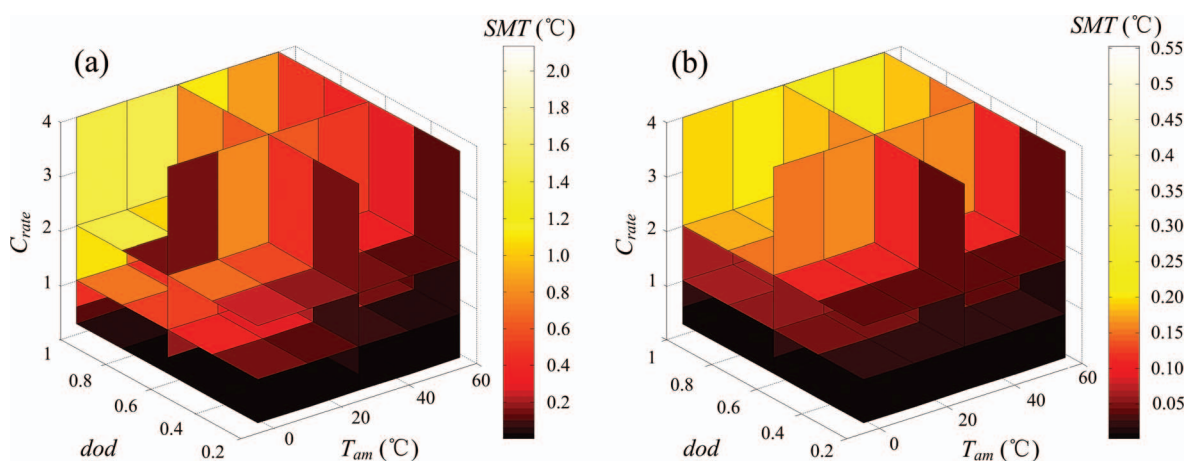


Figure 8.  $T_{sh}$  parameter sensitivity matrix for (a)  $R_{s,a}$  and (b)  $R_{film,a}$ .

extremely complicated and at worst impossible because of the diverse BCIs for all parameters. In order to solve this problem, parameters that have the same or similar BCI should be grouped into clusters, allowing them to be identified together under relatively few sets of operating conditions that render them sensitive and thus readily identified.

The clustering procedure is as follows:

Firstly, the maximum sensitive point which indicates the BCI was determined for both the  $SMU$  and  $SMT$  and each parameter. For example, the maximum sensitive point of parameter  $R_{film,a}$  is (1,5,5) for  $SMU$  and (5,5,5) for  $SMT$ , i.e. the BCI of  $SMU$  is “very low  $dod$ , very high  $T_{am}$  and very high  $C_{rate}$ ” and the BCI of  $SMT$  is “very high  $dod$ , very high  $T_{am}$  and very high  $C_{rate}$ ”.

Secondly, the values of the  $SMU$  and  $SMT$  matrices were normalized, ensuring that all elements were in the range [0,1], and values in three dimensionalities coordinating to the maximum sensitive point were determined to form a vector with 15 elements. This vector is defined as the “BCI characteristic” and characterizes how similar an operating condition is to the BCI of a given parameter. Fig. 9 shows the BCI characteristic of parameter  $R_{film,a}$ . The blue line is the BCI characteristic of  $SMU$ , the green line is that of  $SMT$  and the red line is the average BCI characteristic, which is obtained from averaging the former two with weighting. The weights are the sensitivity of  $U_{app}$  and  $T_{sh}$ , shown as the number of asterisks in Table III.

Thirdly, the Fuzzy C-Means (FCM) method was used for clustering. The average BCI characteristic was used as clustering feature. The “Clustering results” column in Table III shows the degree of association of each parameter with each of the three clusters. The cluster centers are shown in Fig. 10. 13 parameters were allocated to Cluster

A, 7 parameters to Cluster B and 5 parameters to Cluster C, since their membership is greater than 0.5. Two parameters ( $v_0$ ,  $\bar{E}_{k,s,a}$ ) could not be obviously clustered.

The conditions indicated by the highest points of the cluster centers are defined as the “Best Practicable Conditions (BPC)”. Under these conditions the parameters in the same cluster could be at or close to their own BCIs. Thus a simple set of conditions could be used to identify a cluster of parameters, yielding a group of highly identifiable

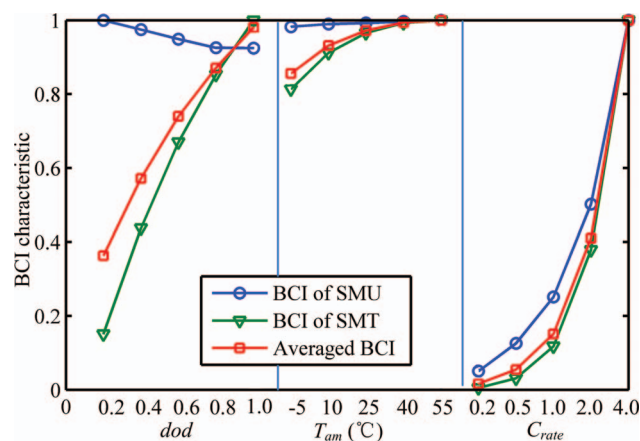


Figure 9. The BCI characteristic for parameter  $R_{film,a}$ .

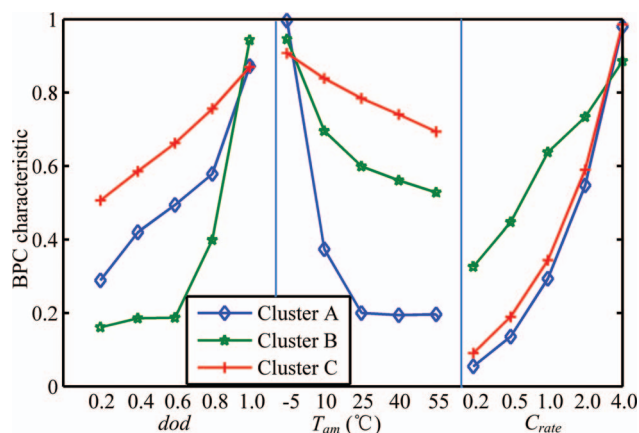


Figure 10. Cluster centers, regarded as BPC characteristics.

parameters. In short, BCI is the condition that makes one parameter relatively most sensitive, and BPC is the condition that makes a cluster of parameters relatively most sensitive synchronously. The “BPC characteristic” shows how similar an operating condition is to a cluster’s BPC.

In Fig. 10, these three BPC characteristics are very similar in the trend, the conditions of higher  $dod$ , lower  $T_{am}$  and higher  $C_{rate}$  are more similar to BPC, while in contrast the conditions of lower  $dod$ , higher  $T_{am}$  and lower  $C_{rate}$  are far away from BPC.

Finally, these three BPC characteristics were daringly combined by averaging. A combined BPC characteristic was obtained as shown in Fig. 11. Theoretically, the combined BPC “very high  $dod$ , very low  $T_{am}$ , and very high  $C_{rate}$ ” would make most of the parameters relatively sensitive, making the experiments much simpler in this case. However, the sensitivity of some parameters may be lower than others due to differences between the combined BPC and their own BCI. Hence, this combined BPC characteristic must be used with caution in the design of identification experiments, and the rationality of such experiments should always be subsequently validated.

### Stepwise Experiment for Parameter Identification

Identifying all parameters synchronously using a simple experiment has some disadvantages. Due to the large number of parameters and the searching space is very large, the optimal solution will be difficult to obtain. In addition, it is computationally expensive to identify all parameters using a parameter optimization algorithm and it is difficult to identify the low sensitive parameters correctly, because the model output is dominated by highly sensitive parameters. Therefore,

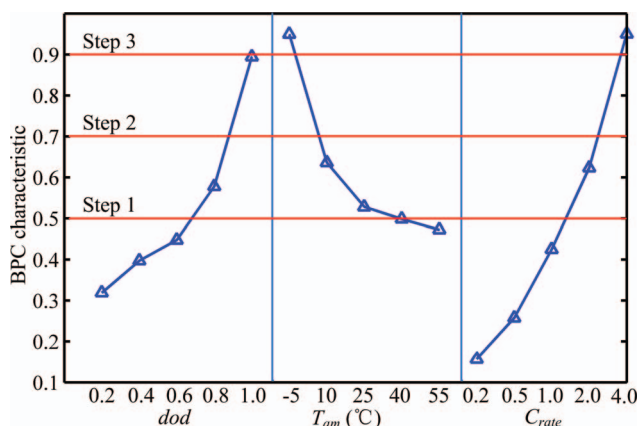


Figure 11. Combined BPC characteristics.

a stepwise identification is proposed based on the results of parameter sensitivity analysis to overcome those disadvantages.

**Experimental design.**— The stepwise experiment should be designed according to the following criteria:

- (1) “Insensitive” parameters should be excluded from identification.
- (2) “Highly sensitive” parameters should be identified first and the results used to identify the “Sensitive” parameters in the second step. Similarly, the results of previous steps are used to identify the “Low sensitive” parameters.
- (3) Operating conditions of the test in each step should refer to “BPC”.
- (4) Tests in the stepwise experiment should be as simple as possible. Conditions that are harmful to the batteries should be avoided.

The major challenge in stepwise identification is that when we identify parameters in the first step, the values of some parameters (to be identified in later steps) are still unknown. Estimated values are input into the model because the model needs a whole parameter set to complete the simulation. The choice of estimated value can influence on model output and thus the identification results may be incorrect. To overcome this problem, the stepwise experiment should be designed using sensitivity grouping results. The key concept is to ensure that the sensitivity of the parameters to be identified is relatively high while that of the other parameters is low. Hence, in Step 1 the “highly sensitive” parameters should be identified under conditions not similar to BPC, in order to ensure low sensitivity of the other parameters while maintaining the identifiability of the “highly sensitive” parameters. Similarly, the conditions used for identification of the “low sensitive” parameters should be similar to BPC, to provide a high identifiability.

Therefore, the following three-step identification was designed based on the BPC characteristic (Fig. 11), taking into account the ease and simplicity of experiments. The details of the experimental scheme are shown in Fig. 12.

Step 1: The operating conditions are  $dod_1 = 0.6$ ,  $T_{am,1} = 30^\circ\text{C}$  and  $C_{rate,1} = 2\text{C}$ . Under these conditions, most parameters have relatively low sensitivity, but highly sensitive parameters should remain sensitive. Benchmark values are used for the “sensitive” and “low sensitive” parameters, listed in Table I as estimated values. Using this strategy, even if the estimated values are far from the true values, the model output will not be significantly affected and highly sensitive parameters will still be identified correctly.

Step 2: The operating conditions are  $dod_2 = 0.8$ ,  $T_{am,2} = 10^\circ\text{C}$  and  $C_{rate,2} = 3\text{C}$ . Under these conditions, parameters are more sensitive

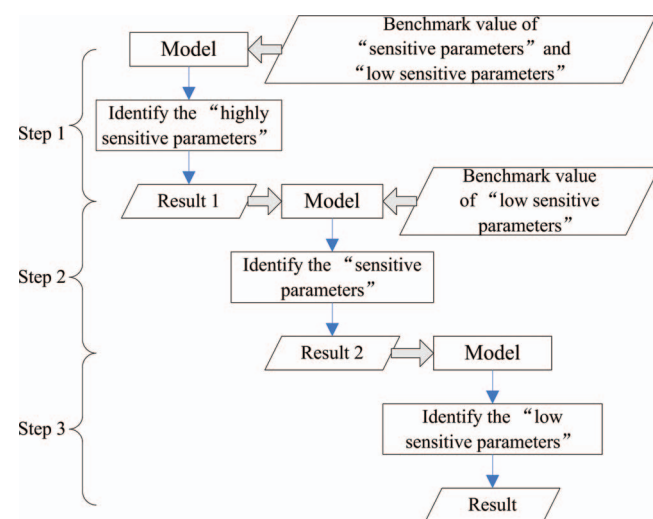


Figure 12. Flow chart of stepwise identification.

**Table IV. Parameter sensitivities in three-step experiment.**

| Parameter           | sens.avg | Step 1   |        |        |        |                | Step 2 |          |     |        |                | Step 3   |        |        |        |                |
|---------------------|----------|----------|--------|--------|--------|----------------|--------|----------|-----|--------|----------------|----------|--------|--------|--------|----------------|
|                     |          | SMU      | sens.U | SMT    | sens.T | sens.identify? | SMU    | sens.U   | SMT | sens.T | sens.identify? | SMU      | sens.U | SMT    | sens.T | sens.identify? |
| $R_{s,a}$           | ***      | 0.0285   | ***    | 0.5789 | ***    | ***            | Y      | 0.0625   |     | 1.1222 |                | 0.0621   |        | 1.3522 |        |                |
| $R_{s,c}$           | **       | 0.0023   | *      | 0.0713 | ***    | **             |        | 0.0053   | **  | 0.2178 | ***            | 0.0147   |        | 0.3392 |        |                |
| $D_{s,a}$           | ***      | 0.0114   | ***    | 0.0911 | **     | ***            | Y      | 0.0241   |     | 0.2580 |                | 0.0557   |        | 0.5045 |        |                |
| $D_{s,c}$           | *        | 0.0009   | /      | 0.0338 | *      | /              |        | 0.0021   | *   | 0.0383 | ***            | 0.0070   |        | 0.1106 |        |                |
| $\sigma_{s,a}$      | /        | 1.48E-05 | /      | 0.0004 | /      | /              |        | 1.90E-05 | /   | 0.0007 | /              | 2.13E-05 | /      | 0.0010 | /      | /              |
| $\sigma_{s,c}$      | **       | 0.0030   | *      | 0.0927 | **     | *              |        | 0.0059   | **  | 0.1608 | **             | 0.0048   |        | 0.2101 |        |                |
| $x_0$               | ***      | 0.0173   | ***    | 0.0737 | **     | ***            | Y      | 0.0195   |     | 0.0839 |                | 0.0235   |        | 0.0957 |        |                |
| $y_0$               | *        | 0.0015   | /      | 0.0518 | **     | *              |        | 0.0019   | /   | 0.0816 | **             | 0.0030   | *      | 0.0986 | **     | *              |
| $\varepsilon_{s,a}$ | ***      | 0.0202   | ***    | 0.0975 | **     | ***            | Y      | 0.0240   |     | 0.1593 |                | 0.0306   |        | 0.2142 |        |                |
| $\varepsilon_{s,c}$ | *        | 0.0014   | /      | 0.0413 | *      | /              |        | 0.0030   | *   | 0.0754 | **             | 0.0048   | *      | 0.1240 | **     | *              |
| $R_{film,a}$        | **       | 0.0049   | *      | 0.1435 | **     | *              |        | 0.0067   | **  | 0.2349 | ***            | 0.0080   |        | 0.3138 |        |                |
| $R_{film,c}$        | /        | 3.17E-05 | /      | 0.0008 | /      | /              |        | 6.27E-05 | /   | 0.0020 | /              | 5.49E-05 | /      | 0.0021 | /      | /              |
| $C_e$               | **       | 0.0027   | *      | 0.0722 | **     | *              |        | 0.0052   | **  | 0.1399 | **             | 0.0052   |        | 0.1891 |        |                |
| $D_e$               | **       | 0.0027   | *      | 0.0733 | **     | *              |        | 0.0055   | **  | 0.1317 | **             | 0.0070   |        | 0.2075 |        |                |
| $\kappa_e$          | ***      | 0.0148   | ***    | 0.4127 | ***    | ***            | Y      | 0.0345   |     | 1.1484 |                | 0.0503   |        | 1.9458 |        |                |
| $\rho_e$            | /        | 3.37E-05 | /      | 0.0151 | /      | /              |        | 0.0002   | /   | 0.0422 | *              | 0.0006   | /      | 0.0499 | *      | /              |
| $R_{ext}$           | ***      | 0.0168   | ***    | 0.4872 | ***    | ***            | Y      | 0.0223   |     | 0.7878 |                | 0.0259   |        | 1.0326 |        |                |
| $\varepsilon_{e,a}$ | ***      | 0.0253   | ***    | 0.0978 | **     | ***            | Y      | 0.0308   |     | 0.1323 |                | 0.0176   |        | 0.0882 |        |                |
| $\varepsilon_{e,s}$ | *        | 0.0007   | /      | 0.0215 | *      | /              |        | 0.0016   | /   | 0.0588 | **             | 0.0051   | **     | 0.0970 | **     | Y              |
| $\varepsilon_{e,c}$ | *        | 0.0009   | /      | 0.0123 | /      | /              |        | 0.0024   | *   | 0.0373 | *              | 0.0064   | **     | 0.0706 | **     | Y              |
| $k_{s,a}$           | ***      | 0.0484   | ***    | 1.3582 | ***    | ***            | Y      | 0.0503   |     | 1.7089 |                | 0.0480   |        | 1.8014 |        |                |
| $k_{s,c}$           | ***      | 0.0146   | ***    | 0.4424 | ***    | ***            | Y      | 0.0229   |     | 0.8554 |                | 0.0251   |        | 1.0518 |        |                |
| $\bar{E}k_{s,a}$    | **       | 0.0030   | *      | 0.0720 | **     | *              |        | 0.0035   | *   | 0.1568 | **             | 0.0065   | **     | 0.3080 | ***    | Y              |
| $\bar{E}k_{s,c}$    | *        | 0.0006   | /      | 0.0153 | /      | /              |        | 0.0017   | /   | 0.0612 | **             | 0.0062   | **     | 0.1566 | **     | Y              |
| $\bar{E}D_{s,a}$    | *        | 0.0007   | /      | 0.0046 | /      | /              |        | 0.0051   | *   | 0.0397 | *              | 0.0137   | ***    | 0.1225 | **     | Y              |
| $\bar{E}D_{s,c}$    | *        | 0.0001   | /      | 0.0038 | /      | /              |        | 0.0009   | /   | 0.0145 | /              | 0.0034   | *      | 0.0693 | **     | Y              |
| $\bar{E}D_e$        | *        | 0.0006   | /      | 0.0153 | /      | /              |        | 0.0010   | /   | 0.0415 | *              | 0.0029   | *      | 0.1375 | **     | Y              |
| $\bar{E}\kappa_e$   | ***      | 0.0038   | *      | 0.0972 | **     | *              |        | 0.0068   | **  | 0.2972 | ***            | 0.0187   |        | 0.8880 |        |                |
| $\lambda$           | *        | 0.0003   | /      | 0.1772 | **     | *              |        | 0.0013   | /   | 0.5258 | ***            | 0.0025   |        | 0.9419 |        |                |
| $h$                 | **       | 0.0009   | /      | 0.6689 | ***    | *              |        | 0.0041   | *   | 1.5329 | ***            | 0.0082   |        | 2.2908 |        |                |

\*\*\*Highly sensitive.

\*\*Sensitive.

\*Low sensitive.

/Insensitive.

than in Step 1, and the “sensitive” parameters can meet the sensitivity demand of identification, while the sensitivity of “low sensitive” parameters is still sufficiently low. In this step, the identified results of “highly sensitive” parameters and benchmark values of “low sensitive” parameters are input into model.

Step 3: The operating conditions are  $dod_3 = 1.0$ ,  $T_{am,3} = 0^\circ\text{C}$  and  $C_{rate,3} = 4\text{C}$ . Under these conditions, all parameters are relatively highly sensitive and the “low sensitive” parameters are now easier to identify. Previously identified values of other parameters are used as model inputs.

It should be noted that the stepwise experiments described above are not universally applicable and should only be used for this particular type of LiFePO<sub>4</sub> battery, whose parameters are listed in Table I and Table II. For other types of battery, it is necessary to obtain benchmark values for parameters by consulting the manufacturer and/or relevant literature, then analyze the parameter sensitivity and design experiments using the proposed methods, rather than using the experiments designed in this paper directly.

**Rationality validation.**— Using the same methodology presented before, the parameters sensitivities were analyzed under the three-step conditions, with the results summarized in Table IV. The “Identify?” column indicates whether the parameter meets the sensitivity demand of identification in a certain step, while a blank space means the parameter has been identified in a previous step and need not be analyzed. The validation results were also used to revise the parameter groupings according to their sensitivity.

The validation results are consistent with the average sensitivity grouping. “Highly sensitive” parameters exhibit high sensitivity in Step 1, while other parameter sensitivities are relatively low (even lower than their average sensitivity). There is a special case in Step 1, the highly sensitive parameter  $\bar{E}\kappa_e$  becomes low sensitive, so it will therefore be identified in a latter step. In Step 2, “sensitive” parameters remain sensitive even become highly sensitive. Two low sensitive parameters  $D_{s,c}$  and  $\lambda$  becomes sensitive, If its estimated value is used, the identification results may be adversely influenced, so these two parameters must now be identified in this step. In Step 3, most “low sensitive” parameters become “sensitive” or even “highly sensitive”, so they can be identified more accurately, but there are still three parameters show “low sensitive”, implying that their identification accuracy may be lower than the others.

There are 9 parameters in each step in the revised stepwise experiment, indicated by “Y” in the “Identify?” column in Table IV. The parameters are relatively sensitive when they are identified, which means that their identifiability is high. The parameters with estimated values before identification show relatively low sensitivity and have very little influence on model output. Therefore we can conclude that this stepwise experiment is rationally designed.

**Effectiveness of stepwise experiment for parameter identification.**— Synthetic experimental data is used to verify the effectiveness of the stepwise experiment. This synthetic experimental data is not derived from real experiments but from simulation, thus it has many advantages:

Table V. Results of stepwise identification.

| Parameter                                                         | value    | Loop 1   |        | Loop 2   |        |
|-------------------------------------------------------------------|----------|----------|--------|----------|--------|
|                                                                   |          | result   | error% | result   | error% |
| $R_{s,a}$ (m)                                                     | 3.60e-06 | 3.41e-06 | 5.26%  | 3.36e-06 | 6.54%  |
| $R_{s,c}$ (m)                                                     | 4.00e-08 | 3.75e-08 | 6.32%  | 3.70e-08 | 7.59%  |
| $D_{s,a}$ (m <sup>2</sup> s <sup>-1</sup> )                       | 4.00e-14 | 3.68e-14 | 7.89%  | 3.84e-14 | 3.99%  |
| $D_{s,c}$ (m <sup>2</sup> s <sup>-1</sup> )                       | 3.00e-18 | 2.89e-18 | 3.57%  | 3.05e-18 | 1.57%  |
| $\sigma_{s,c}$ (S m <sup>-1</sup> )                               | 0.6      | 0.6444   | 7.41%  | 0.6361   | 6.03%  |
| $x_0$ (-)                                                         | 0.81     | 0.8135   | 0.43%  | 0.8145   | 0.56%  |
| $y_0$ (-)                                                         | 0.05     | 0.0498   | 0.38%  | 0.0495   | 0.96%  |
| $\varepsilon_{s,a}$ (-)                                           | 0.6      | 0.6187   | 3.12%  | 0.6112   | 1.88%  |
| $\varepsilon_{s,c}$ (-)                                           | 0.44     | 0.4513   | 2.57%  | 0.4669   | 6.12%  |
| $R_{film,a}$ ( $\Omega$ m <sup>2</sup> )                          | 0.004    | 0.0033   | 16.96% | 0.0036   | 9.01%  |
| $C_e$ (mol m <sup>-3</sup> )                                      | 1100     | 1072.22  | 2.52%  | 1079.83  | 1.83%  |
| $D_e$ (m <sup>2</sup> s <sup>-1</sup> )                           | 4.50e-10 | 4.72e-10 | 4.91%  | 4.61e-10 | 2.53%  |
| $\kappa_e$ (S m <sup>-1</sup> )                                   | 0.8      | 0.8085   | 1.06%  | 0.8459   | 5.74%  |
| $R_{ext}$ ( $\Omega$ m <sup>2</sup> )                             | 0.002    | 0.0020   | 2.91%  | 0.0020   | 3.13%  |
| $\varepsilon_{e,a}$ (-)                                           | 0.35     | 0.3513   | 0.39%  | 0.3352   | 4.20%  |
| $\varepsilon_{e,s}$ (-)                                           | 0.5      | 0.5339   | 6.80%  | 0.5506   | 10.12% |
| $\varepsilon_{e,c}$ (-)                                           | 0.35     | 0.3468   | 0.89%  | 0.3408   | 2.62%  |
| $k_{s,a}$ (m <sup>2.5</sup> mol <sup>-0.5</sup> s <sup>-1</sup> ) | 5.00e-11 | 4.59e-11 | 8.20%  | 4.75e-11 | 4.96%  |
| $k_{s,c}$ (m <sup>2.5</sup> mol <sup>-0.5</sup> s <sup>-1</sup> ) | 3.00e-12 | 2.44e-12 | 18.55% | 2.69e-12 | 10.34% |
| $\bar{E}k_{s,a}$ (KJ mol <sup>-1</sup> )                          | 25       | 25.0187  | 0.07%  | 25.1146  | 0.46%  |
| $\bar{E}k_{s,c}$ (KJ mol <sup>-1</sup> )                          | 28       | 34.0040  | 21.44% | 33.8409  | 20.86% |
| $\bar{E}D_{s,a}$ (KJ mol <sup>-1</sup> )                          | 32       | 34.2830  | 7.13%  | 34.5551  | 7.98%  |
| $\bar{E}D_{s,c}$ (KJ mol <sup>-1</sup> )                          | 30       | 29.1818  | 2.73%  | 28.5648  | 4.78%  |
| $\bar{E}D_e$ (KJ mol <sup>-1</sup> )                              | 14       | 13.8902  | 0.78%  | 13.4983  | 3.58%  |
| $\bar{E}\kappa_e$ (KJ mol <sup>-1</sup> )                         | 23.5     | 28.4294  | 20.98% | 27.7547  | 18.11% |
| $\lambda$ (W m <sup>-1</sup> K <sup>-1</sup> )                    | 0.7      | 0.7063   | 0.90%  | 0.7163   | 2.33%  |
| $h$ (W m <sup>-2</sup> K <sup>-1</sup> )                          | 25       | 25.1078  | 0.43%  | 24.9191  | 0.32%  |
| average error%                                                    |          |          | 5.73%  |          | 5.49%  |

- (1) The “true values” of parameters can be treated as known, because we set them ourselves and use them to acquire synthetic experimental data via simulation. The identification results are compared with the “true values,” so the identified error can then be calculated.
- (2) The use of synthetic experimental data circumvents the issues of cost, delay and measurement error associated with experimental test programs.

In order to assess the effectiveness of the stepwise experiment, the “true values” of 27 parameters were set, taking into account the typical range and benchmark values. The set parameter values are shown in the “Value” column in Table V. These parameters were used for model simulation under the operating conditions obtained in Section 4.1 and the results are regarded as synthetic experimental data. Finally, a weight-based multi-objective Genetic Algorithm (GA) was used for parameter identification, in which the terminal voltage  $U_{app}$  and surface temperature  $T_{sh}$  were treated as two objects. The object function is the sum of squares error (SSE) of the whole simulated curve and synthetic experimental data. A lower SSE means the simulated curve fits the synthetic experimental data better and the identified results are closer to the “true values.”

During the identification work, we found that the identification accuracy using the particular *dod* region and that using the whole discharge curve are almost the same, and in the three-step experiments, the battery needs to be fully charge/discharge for three times, the whole curves can be acquired as easily as the particular *dod* curves in the experiments. What is more interesting, the whole discharge curve includes the initial terminal voltage, which determined by the  $E_{ocv}$  of the cathode and the parameter  $y_0$ . Parameter  $y_0$  shows sensitive only in the very low *dod* region, and in other *dod* regions it is low sensitive or insensitive. The sensitivity value matrix of  $y_0$  is very different from other parameters, making it failed to be clustered in any cluster, so it shows low sensitive in the designed stepwise experiments, proved in Table IV. The identification accuracy of  $y_0$  will be higher by using the

whole discharge curves. So, in this section the whole discharge curve is used to calculate the objective function instead of the particular *dod* region.

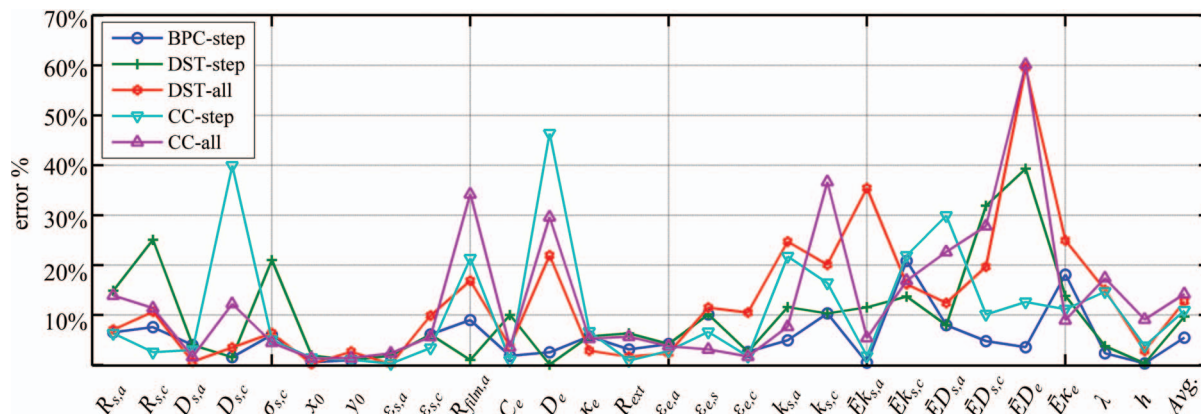
The Genetic Algorithm Toolbox<sup>24</sup> was used to program the parameter identification in MATLAB. Since there is only one test in each step, a single population GA was used. For cases in which there is more than one test in each step due to complex BPC, the Cooperative Co-evolutionary Genetic Algorithm (CCGA) should be used, and the population partition should be set according to the parameter clustering results. The model simulation program mentioned in Section 2.3 was called  $3 \times N_i \times MAXGEN$  times during the whole identification process, which is computationally expensive and takes more than one week to complete. In order to speed up the identification, a computing cluster with 20 cores was built using an HP ProLiant DL160 G6 server (2×Intel L5520 @ 2.26GHz CPU, 16G RAM) and five PCs (Intel i3–530 @ 2.93GHz CPU, 4G RAM) based on the MATLAB Distributed Computing Toolbox. This resulted in a 14-fold increase in the speed of the GA identification process according to our previous work.<sup>25</sup> Table VI shows the preferences of GA for the three steps, where  $N_d$  is the number of variables,  $N_i$  is the size of population,  $MAXGEN$  is the maximum generation,  $GGAP$  is the generation gap,  $INSR$  is the insert rate and  $P_m$  is the mutation probability. For details of GA-based parameter identification, the reader is referred to Ref. 26.

Some changes were made to Step 2 and Step 3 during this process. The values of identified parameters were not used directly, but

Table VI. Preferences of Genetic Algorithm in 3-step parameter identification.

| STEP | $N_d$ | $N_i$ | MAXGEN | GGAP | INSR | $P_m$ |
|------|-------|-------|--------|------|------|-------|
| 1    | 9     | 600   | 50     | 0.8  | 0.8  | 1/9   |
| 2    | 18    | 600   | 50     | 0.8  | 0.8  | 1/18  |
| 3    | 27    | 600   | 50     | 0.8  | 0.8  | 1/27  |





**Figure 13.** Comparison of the relative errors in parameter identification using different tests.

were identified again over a smaller range  $[lower', upper']$  defined as follows:

$$\begin{cases} lower' = lower + (value - lower) \times 0.5 \\ upper' = upper - (upper - value) \times 0.5 \end{cases} \quad [11]$$

where *value* is the identified result in previous step and *lower* and *upper* represent the searching range in the previous step. This was deemed necessary as although the operating conditions in Step 1 can decrease parameter sensitivity and reduce the influence of estimated values on model output, it cannot guarantee the accurate identification of “highly sensitive” parameters. The values of “highly sensitive” parameters are very close to the true values, which affects the model output slightly, thus “sensitive” parameters still remain highly identifiable. In the same manner, “highly sensitive” parameters and “sensitive” parameters were identified over a smaller range in Step 3.

In order to improve the accuracy, another loop from step 1 to step 3 was conducted. In Step 1 Loop 2, the identified values of the sensitive parameters and the low sensitive parameters which obtained in Loop 1 were used to complete the identification of highly sensitive parameters; and in Step 2 Loop 2, the identified low sensitive parameter values which obtained in Loop 1 are used. The identified results are shown in Table V, and the Column “error%” gives the relative errors between “true value” and identified result.

It is seen that, the identification accuracy of Loop 2 is improved but not significantly. We speculate the possible reasons may be: (1) The multi-physics model is highly non-linear, the relationship of the model output and the parameters is extremely complex. When the guessed values of low sensitive parameters are close to the “true value”, small errors of them almost cannot affect the output curves, thus the results of Loop 1 are accurate enough. (2) The Genetic Algorithm is one of the stochastic algorithms, random number is used in the recombination operator and the mutation to generate the solution, so it is very hard even impossible to obtain the absolutely accurate parameter values. Therefore, the identification results of Loop 2 are hardly to be improved obviously. Besides the GA, other algorithms are encouraged to be used to identify the parameters with the stepwise experiments.

The verification process was carried out as a single blind test in that the set “true values” were unknown to the researcher carrying out the simulation. In this way the use of synthetic experimental data in the identification process is very similar to testing a real battery. The identified results using synthetic experimental data are shown in Table V, and the “error%” column lists the relative error between the “true value” and the identified result.

As a comparison, a Dynamic Stress Test (DST) and a constant current (CC) test at room temperature were also used for parameter identification. Fig. 13 shows the comparison of the relative error of each parameter in different tests, and the Avg. stands for the average error of all parameters in a certain test. In each test, the parameters are

identified both stepwise and synchronously (respectively expressed by “-step” and “-all” in the legend).

In general, identification errors using the stepwise BPC test are lower than others. The highest errors were observed for  $\bar{E}k_{s,c}$  (20.86%) and  $\bar{E}k_e$  (18.11%) because their BCI is far away from the combined BPC, making these two parameters “low sensitive” when identified. This is demonstrated clearly in the “Clustering results” column in Table III,  $\bar{E}k_{s,c}$  ambiguously belong to Cluster A and  $\bar{E}k_e$  does not belong to any cluster due to small memberships.

The identification accuracy using the stepwise DST test is also high, albeit lower than that using BPC test. Though there are many current levels included in the DST, there is only one ambient temperature condition and as a result some “low sensitive” parameters related to temperature (e.g. activation energy) are unsatisfactorily identified. The CC test contains only one current and one ambient temperature. It is far away from the BPC, and therefore yields high errors for “low sensitive” parameters.

The “low sensitive” and “sensitive” parameters such as  $D_{s,c}$ ,  $R_{fil,m}$ ,  $D_e$  and activation energies are unsatisfactorily identified using the DST or CC test when identified together, because they have very little influence on the terminal voltage and surface temperature compared to “highly sensitive” parameters. As a result, the accuracy in identifying all 27 parameters synchronously is much lower than that using a stepwise experiment. The stepwise experiment designed from BPC described in this article clearly demonstrates better performance than others in identification accuracy.

It is interesting that although the parameters  $y_0$  and  $\varepsilon_{s,c}$  are "low sensitive" they can still be identified accurately in all tests, because their values are directly related to the battery capacity and initial terminal voltage. This special case shows that there remains much work to do on understanding parameter sensitivity analysis, particularly with regard to experiment design.

## Conclusions

This article has presented a parameter sensitivity analysis of a cylindrical Li-ion battery based on a multi-physics model, using a carefully designed stepwise experiment. The main conclusions are as follows:

- (1) A multi-physics model of a cylindrical Li-ion battery has been developed by coupling a thermal distribution model to an electrochemical P2D model, allowing simulation of the surface temperature, which can be easily validated through measurement. 30 parameters are identified in this model, including many which can be used to characterize battery health.
- (2) The sensitivity of terminal voltage  $U_{app}$  and surface temperature  $T_{sh}$  to these parameters were examined using model simulation under 25 different conditions. A parameter sensitivity matrix was proposed to describe the variation of  $U_{app}$  and  $T_{sh}$  curves

with 5 different parameter values under a range of conditions and the 30 parameters were then roughly grouped according to their average and maximum sensitivity. 10 “highly sensitive”, 7 “sensitive” and 10 “low sensitive” parameters were identified. Three “insensitive” parameters were excluded from further analysis as they cannot be used to study PHM characteristics.

- (3) The sensitivity of a parameter is strongly dependent on operating conditions. A BCI characteristic was proposed to indicate the operating conditions under which a parameter has the highest sensitivity. The best identification strategy is to identify each parameter under its own BCI or under conditions similar to its BCI. The parameters were grouped into 3 clusters based on their BCI characteristic using the Fuzzy C-mean Method, with the clustering center defined as BPC, which is defined as the conditions in which a cluster of parameters are synchronously most sensitive.
- (4) Three BPC characteristics were combined because they are similar in the trend and the combined BPC characteristic could simplify experiment design. A three-step experiment was designed for a 2.3 Ah LiFePO<sub>4</sub> battery. In each step, 9 parameters were identified under a simple set of operating conditions. The rationality was validated using model simulation under each set of conditions. The identification accuracy of stepwise BPC, DST and CC conditions was compared using synthetic experimental data. The stepwise BPC showed the highest accuracy with an average relative error of 5.49%, while the accuracy of identifying all 27 parameters synchronously was also higher.

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## Appendix A. Electrochemical Model

### Faraday's laws

The electrochemical reaction current density is used to describe the flux of Li<sup>+</sup> on the intercalation particle of the electrode:

$$i_s = nFj_{Li} \quad [A1]$$

where  $n = 1$  is the charge number of the lithium ion (all variables are described in the Nomenclature section).

The ionic current density in the solution phase modulation is determined by:

$$\nabla \cdot i_2 = a_s i_s = a_s F j_{Li} \quad [A2]$$

where  $a_s$  is specific surface area of active material particles defined as  $a_s = \frac{3\epsilon_s}{R_s}$  and the electron current density in the solid phase is:

$$\nabla \cdot i_1 = -a_s i_s = -a_s F j_{Li} \quad [A3]$$

The applied current is equal to the product of electron current density at boundary 1 or 4 and the electrode area:

$$i_{app} = A_{cell} i_1|_1 = A_{cell} i_1|_4 \quad [A4]$$

The total current density inside the battery is given by the sum of electron current density and ionic current density:

$$i = i_1 + i_2 \quad [A5]$$

### Charge balance and Ohm's law

In active materials the solid phase potential is calculated from Ohm's law:

$$i_1 = -\sigma_s^{eff} \nabla \phi_s \quad [A6]$$

where the effective conductivity of solid phase  $\sigma_s^{eff} = \sigma_s \epsilon_s^{1.5}$ ,  $\phi_s$  is 0 at boundary 1 and the charge flux is zero at boundaries 2 and 3:

$$\begin{cases} \phi_s|_1 = 0 \\ \left. \frac{\partial \phi_s}{\partial x} \right|_4 = -\frac{i}{\sigma_s^{eff}} \end{cases} \quad [A7]$$

$$\left. \frac{\partial \phi_s}{\partial x} \right|_{2-} = \left. \frac{\partial \phi_s}{\partial x} \right|_{3+} = 0 \quad [A8]$$

In the electrolyte, the solution phase potential is determined by Ohm's law and ionic transport of charges:

$$i_2 = -\kappa_e^{eff} \nabla \phi_e + \frac{2\kappa_e^{eff} RT}{F} \left( 1 + \frac{\partial \ln f_{\pm}}{\partial \ln C_e} \right) (1 - t_+^0) \nabla \ln C_e \quad [A9]$$

Liquid-junction potential is introduced with the expression:

$$K_{junc} = \frac{2RT}{F} \left( 1 + \frac{\partial \ln f_{\pm}}{\partial \ln C_e} \right) (1 - t_+^0) \quad [A10]$$

and  $\kappa_e^{eff} = \kappa_e \epsilon_e^{1.5}$  is the effective conductivity of the solution phase.  $\phi_e$  is taken to be continuous at boundaries 2 and 3:

$$\begin{cases} \phi_e|_{2-} = \phi_e|_{2+} \\ \phi_e|_{3-} = \phi_e|_{3+} \end{cases} \quad [A11]$$

while it is set to zero at boundaries 1 and 4:

$$\left. \frac{\partial \phi_e}{\partial x} \right|_1 = \left. \frac{\partial \phi_e}{\partial x} \right|_4 = 0 \quad [A12]$$

### Mass balance and transport process

The mass balance of Li<sup>+</sup> in a spherical particle of active material is described by Fick's law:

$$\frac{\partial C_s}{\partial t} = \frac{1}{r^2} \nabla \cdot (D_s r^2 \nabla C_s) \quad [A13]$$

where  $r$  is the distance from the center of the particle. The lithium concentration at the surface is coupled to  $C_{e,s}$  and the reaction current density, while the Li<sup>+</sup> flux is set to zero at the center of the particle, because there is no species source:

$$\begin{cases} \left. \frac{\partial C_s}{\partial r} \right|_{r=R_s} = -\frac{j_{Li}}{D_s} \\ \left. \frac{\partial C_s}{\partial r} \right|_{r=0} = 0 \end{cases} \quad [A14]$$

The Li<sup>+</sup> concentration in solution, which varies due to diffusion and reaction at the electrode/electrolyte interface, is calculated from the following equation:

$$\epsilon_e \frac{\partial C_e}{\partial t} = \nabla \cdot (D_e^{eff} \nabla C_e) + \frac{a_s}{F} (1 - t_+^0) \cdot i_s \quad [A15]$$

Diffusion takes place in three regions (anode, separator and cathode). Current collectors present an impermeable wall to the electrolyte, so the Li<sup>+</sup> flux is zero at external boundaries 1 and 4 and shows a continuity condition at inner boundaries 2 and 3.

$$\begin{cases} \left. \frac{\partial C_e}{\partial x} \right|_1 = \left. \frac{\partial C_e}{\partial x} \right|_4 = 0 \\ -D_{e,a}^{eff} \left. \frac{\partial C_e}{\partial x} \right|_{2-} = -D_{e,s}^{eff} \left. \frac{\partial C_e}{\partial x} \right|_{2+} \\ -D_{e,s}^{eff} \left. \frac{\partial C_e}{\partial x} \right|_{3-} = -D_{e,c}^{eff} \left. \frac{\partial C_e}{\partial x} \right|_{3+} \end{cases} \quad [A16]$$

At the initial time, the electrolyte concentration is equal to a constant value  $C_{e,0}$ .

### Electrochemical kinetics

The electrochemical reaction occurring at the surface of the active material is described by the Butler-Volmer equation:

$$i_s = nFj_{Li} = i_0 \left[ \exp \left( \frac{\alpha_a F}{RT} \eta \right) - \exp \left( -\frac{\alpha_c F}{RT} \eta \right) \right] \quad [A17]$$

where  $\alpha_a$  and  $\alpha_c$  are the transfer coefficients of anode and cathode respectively and  $\eta$  is the overpotential, which is defined as:

$$\eta = \phi_s - \phi_e - E_{ocv} - i_s R_{film} \quad [A18]$$

$R_{film}$  is the resistance of the solid electrolyte interface (SEI) film and  $E_{ocv}$  is the open circuit potential, which is a function of the solid phase Li<sup>+</sup> concentration at the particle surface. The  $E_{ocv}$  of the MCMB/LiFePO<sub>4</sub> system is given by:<sup>20</sup>

$$\begin{aligned} E_{ocv,a} = & 0.6379 + 0.5416 \exp(-305.5309x) + 0.044 \tanh \left( -\frac{x - 0.1958}{0.1088} \right) \\ & - 0.1978 \tanh \left( \frac{x - 1.0571}{0.0854} \right) - 0.6875 \tanh \left( \frac{x + 0.0117}{0.0529} \right) \\ & - 0.0175 \tanh \left( \frac{x - 0.5692}{0.0875} \right) \end{aligned} \quad [A19]$$

$$E_{ocv,c} = 3.4323 - 0.8428 \exp(-80.2493(1-y)^{1.3198}) - 3.2474 \times 10^{-6} \exp(20.2645(1-y)^{3.8003}) + 3.2482 \times 10^{-6} \exp(20.2646(1-y)^{3.7995}) \quad [A20]$$

where  $x = \frac{C_{e/s,n}}{C_{s,MAX}^{MAX}}$  is the stoichiometric number for the anode and  $y = \frac{C_{e/s,p}}{C_{s,MAX}^{MAX}}$  is the stoichiometric number for the cathode. The exchange current density  $i_0$  acts as a bridge connecting concentrations in both solid and liquid phases, and indicates the ease of electrode polarization. It may be calculated as follows:

$$i_0 = k_s C_e^{a_a} (C_{s,MAX} - C_{e/s})^{a_a} C_{e/s}^{a_c} \quad [A21]$$

where  $k_s$  is the electrochemical reaction rate. All of the above equations should be solved by a numerical method, after which the terminal voltage of the battery is calculated by:

$$U_{app} = \phi_s|_4 - \phi_s|_1 - iR_{ext} \quad [A22]$$

where  $R_{ext}$  is the contact resistance between electrode and current collector.

## Appendix B. Thermal model

### Energy balance of the battery

The energy balance equation contains three terms, namely the heat capacity, the heat conduction and the heat generation, respectively.

$$\rho C_p \frac{\partial T}{\partial t} = \nabla(\lambda \nabla T) + \dot{Q} \quad [B1]$$

where  $\rho$  is the volume-averaged density of the battery,  $C_p$  is the volume-averaged heat capacity and  $\lambda$  is the effective thermal conductivity.

### Heat generation

The heat generation rate is defined by  $\dot{Q} = \dot{Q}_{rea} + \dot{Q}_{ent} + \dot{Q}_{ohm}$ , comprising electrochemical reaction heat, entropic heat and ohmic heat. These are calculated using following equations. Electrochemical reaction heat:

$$\dot{Q}_{rea} = a_s i_s (\phi_s - \phi_e - E_{ocv}) \quad [B2]$$

Entropic heat:

$$\dot{Q}_{ent} = a_s i_s T \frac{dE_{ocv}}{dT} \quad [B3]$$

where  $\frac{dE_{ocv}}{dT}$  is the entropy coefficient of the electrode, related to the stoichiometric number. The entropy coefficient of the MCMB/LiFePO<sub>4</sub> system can be obtained from Ref. 18.

Ohmic heat can be divided into three parts:

$$\dot{Q}_{ohm} = \dot{Q}_{ohm,s} + \dot{Q}_{ohm,e} + \dot{Q}_{ext} \quad [B4]$$

The first term represents solid phase ohmic heat, generated by the electron current in the active material, the second term is the sum of solution phase ohmic heat and ionic migration heat and the third term is the ohmic heat associated with contact resistance:

$$\begin{cases} \dot{Q}_{ohm,s} = \sigma_s^{eff} \nabla \phi_s \nabla \phi_s \\ \dot{Q}_{ohm,e} = \kappa_e^{eff} \nabla \phi_e \nabla \phi_e + \kappa_e^{D,eff} \nabla \ln C_e \nabla \phi_e \\ \dot{Q}_{ext} = i_{app}^2 R_{ext} / L \end{cases} \quad [B5]$$

where  $L$  is the total thickness of the battery and  $\kappa_e^{D,eff} = \frac{2RT\kappa_e^{eff}}{F} (t_+^0 - 1)$ .

### Heat exchange

Heat exchange between the battery and ambient is governed by convection and radiation. Heat convection is expressed by Newton's cooling law:

$$\dot{q}_c = h(T_{sh} - T_a) \quad [B6]$$

while heat radiation is expressed by the Stefan-Boltzmann equation:

$$\dot{q}_r = \varepsilon \sigma (T_{sh}^4 - T_a^4) \quad [B7]$$

where  $h$  is the heat transfer coefficient,  $\varepsilon$  is the emissivity,  $\sigma$  is the Stefan-Boltzmann constant,  $T_{sh}$  is the shell temperature of the battery and  $T_a$  is ambient temperature.

### Temperature dependent parameters

Some parameters in the P2D model, such as the solid phase diffusion coefficients,  $D_{s,a}$  and  $D_{s,c}$ , solution phase conductivity,  $\kappa_e$ , solution phase diffusion coefficient,  $D_e$ , and electrochemical reaction rates,  $k_{s,a}$  and  $k_{s,c}$ , are coupled with the battery temperature. These parameters can be updated using the Arrhenius law:

$$X_i = X_{i,ref} \exp \left[ \frac{\bar{E}_i}{R} \left( \frac{1}{T_{ref}} - \frac{1}{T} \right) \right] \quad [B8]$$

where  $X_i$  represents the parameters mentioned above,  $X_{i,ref}$  is the parameter value at the reference temperature,  $T_{ref}$ , and  $\bar{E}_i$  denotes the corresponding activation energy of parameter  $X_i$ . In addition, the open circuit potential depends on battery temperature according to the Nernst equation:

$$E_{ocv} = E_{ocv}^{ref} + (T - T_{ref}) \frac{dE_{ocv}}{dT} \quad [B9]$$

where  $E_{ocv}^{ref}$  is the open circuit potential at the reference temperature.

## List of Symbols

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| $a_s$ :       | specific surface area (m <sup>-1</sup> )                                                                  |
| $A_{cell}$ :  | area of electrode (m <sup>2</sup> )                                                                       |
| $C_e$ :       | electrolyte concentration (mol m <sup>-3</sup> )                                                          |
| $C_s$ :       | concentration of Li <sup>+</sup> in the intercalation particle (mol m <sup>-3</sup> )                     |
| $C_{e/s}$ :   | concentration of Li <sup>+</sup> at the surface of intercalation particle (mol m <sup>-3</sup> )          |
| $C_p$ :       | specific heat capacity (J kg <sup>-1</sup> K <sup>-1</sup> )                                              |
| $D_e$ :       | electrolyte diffusion coefficient (m <sup>2</sup> s <sup>-1</sup> )                                       |
| $D_s$ :       | Li <sup>+</sup> diffusion coefficient in active material (m <sup>2</sup> s <sup>-1</sup> )                |
| $\bar{E}$ :   | activation energy (kJ mol <sup>-1</sup> )                                                                 |
| $E_{ocv}$ :   | open circuit potential (V)                                                                                |
| $F$ :         | Faraday's constant (= 96487 C mol <sup>-1</sup> )                                                         |
| $h$ :         | heat transfer coefficient (W m <sup>-2</sup> K <sup>-1</sup> )                                            |
| $i$ :         | total current density in electrodes or separator (A m <sup>-2</sup> )                                     |
| $i_{app}$ :   | applied current (A)                                                                                       |
| $i_0$ :       | exchange current density (A m <sup>-2</sup> )                                                             |
| $i_1$ :       | solid phase current density (A m <sup>-2</sup> )                                                          |
| $i_2$ :       | solution phase current density (A m <sup>-2</sup> )                                                       |
| $i_s$ :       | electrochemical reaction current density (A m <sup>-2</sup> )                                             |
| $j_{Li}$ :    | flux of Li <sup>+</sup> on the intercalation particle of electrode (mol m <sup>-2</sup> s <sup>-1</sup> ) |
| $k_s$ :       | electrochemical reaction rate (m <sup>2.5</sup> mol <sup>-0.5</sup> s <sup>-1</sup> )                     |
| $L$ :         | thickness of electrode or separator (m)                                                                   |
| $\dot{q}$ :   | heat exchange rate (W m <sup>-2</sup> )                                                                   |
| $\dot{q}_c$ : | heat convection rate (W m <sup>-2</sup> )                                                                 |
| $\dot{q}_r$ : | heat radiation rate (W m <sup>-2</sup> )                                                                  |
| $\dot{Q}$ :   | heat generation rate (W m <sup>-3</sup> )                                                                 |
| $R$ :         | ideal gas constant (= 8.3143 J mol <sup>-1</sup> K <sup>-1</sup> )                                        |
| $R_{ext}$ :   | contact resistance (Ω m <sup>2</sup> )                                                                    |
| $R_{film}$ :  | SEI film resistance (Ω m <sup>2</sup> )                                                                   |
| $R_p$ :       | intercalation particle radius of electrode (m)                                                            |
| $t_+^0$ :     | Li <sup>+</sup> transference number in the electrolyte                                                    |
| $T$ :         | battery temperature (K)                                                                                   |
| $T_a$ :       | ambient temperature (K)                                                                                   |
| $U_{app}$ :   | applied potential (V)                                                                                     |
| $x$ :         | stoichiometric number of anode                                                                            |
| $y$ :         | stoichiometric number of cathode                                                                          |

## Greek symbols

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| $\alpha$ :        | transfer coefficient of the electrochemical reactions                                     |
| $\delta$ :        | thickness of current collector (m)                                                        |
| $\varepsilon$ :   | emissivity                                                                                |
| $\varepsilon_e$ : | volume fraction of electrolyte in electrode or separator                                  |
| $\varepsilon_s$ : | volume fraction of active material in electrode                                           |
| $\eta$ :          | overpotential (V)                                                                         |
| $\kappa_e$ :      | ionic conductivity of electrolyte (S m <sup>-1</sup> )                                    |
| $\lambda$ :       | thermal conductivity (W m <sup>-1</sup> K <sup>-1</sup> )                                 |
| $\rho$ :          | density (kg m <sup>-3</sup> )                                                             |
| $\sigma$ :        | Stefan-Boltzman constant (= 5.6704 × 10 <sup>-8</sup> W m <sup>-2</sup> K <sup>-4</sup> ) |
| $\sigma_s$ :      | solid phase conductivity (S m <sup>-1</sup> )                                             |
| $\phi_s$ :        | solid phase potential (V)                                                                 |
| $\phi_e$ :        | solution phase potential (V)                                                              |

## Subscript

|       |         |
|-------|---------|
| $a$ : | anode   |
| $c$ : | cathode |

s: separator  
 0: initial value  
 MAX: maximum value  
 ref: reference value

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